

EMDA and CFD in Numerical Relativity

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ABSTRACT

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[The abstract is a summary of the thesis/dissertation with emphasis on the findings of the study. The abstract must not exceed 350 words in length and fit on one page, single spaced.]

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Chapter 1

Introduction

1.1 Introduction

Since Einstein introduced the theory of general relativity in 1915 [1], it has remained consistent with all experimental tests to date. General relativity has fundamentally transformed our understanding of space and time, explaining phenomena ranging from the anomalous precession of Mercury to the operation of atomic clocks in GPS satellites, and predicting the existence of compact objects such as neutron stars and black holes, as well as gravitational waves [2, 3].

Gravitational waves were first directly detected by LIGO in 2015 [4]. These signals originate from some of the most energetic events in the universe, such as mergers of neutron stars and black holes. However, the signals detected by LIGO are extremely weak; the detectors are sensitive enough to measure relative length changes on the order of 10^{-21} , corresponding to displacements smaller than a proton diameter [5]. Despite this extraordinary sensitivity, gravitational wave signals remain weak relative to background noise. The highest signal-to-noise ratio (SNR) observed to date is on the order of ~ 80 [6].

Interpreting gravitational wave detections requires accurate theoretical models of the expected signals. In practice, this is achieved by comparing observational data to waveform templates generated through numerical simulations. This comparison allows us to statistically determine key physical properties of the source, such as masses and spins.

Making theoretical models of binary black hole mergers is not trivial. In GR, analytic solutions in general relativity are typically restricted to highly symmetric systems and cannot describe the full dynamics of binary black hole mergers. As a result, numerical relativity simulations are essential. A number of codes have been developed for this purpose, including the Einstein Toolkit, SpEC, GRChombo, BAM, GR-Athena++, and Dendro-GR. [7–12]

Simulating binary black hole mergers is computationally demanding. It requires solving a coupled system of nonlinear partial differential equations across multiple length scales, including

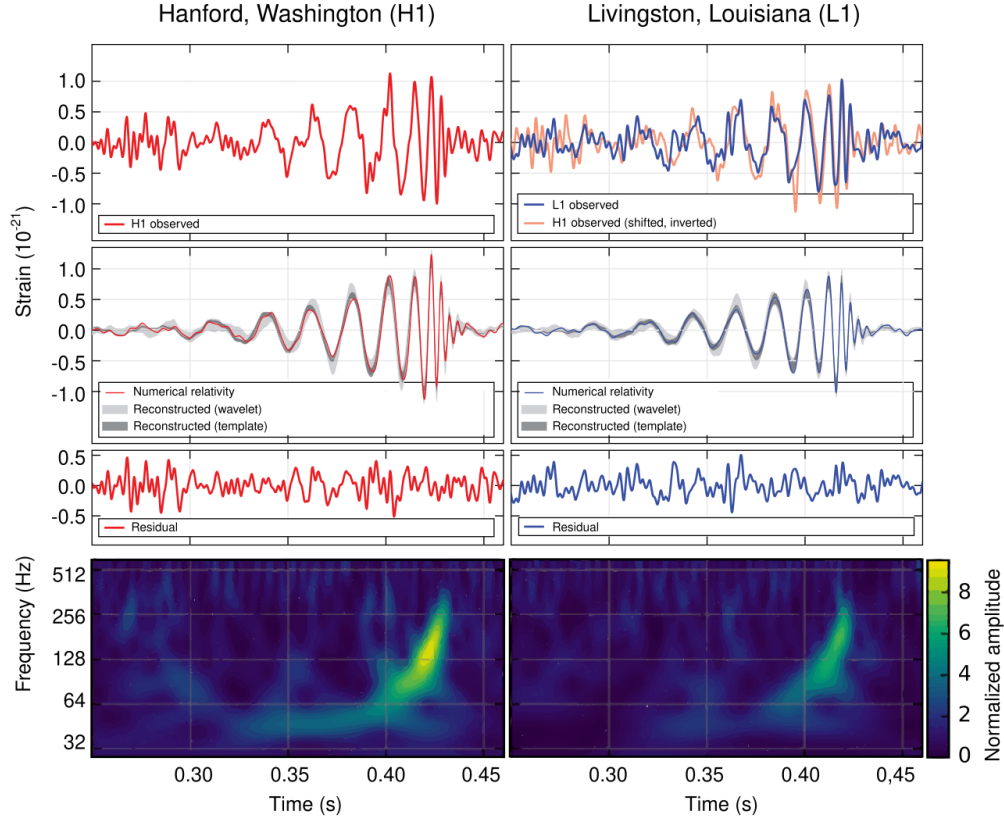


Figure 1.1 Schematic illustration of the LIGO interferometer used to detect gravitational waves through differential arm length changes.

the scale of the black holes, the binary orbital scale, and the radiation zone [13, 14]. In this work, we make use of the Dendro numerical relativity framework [12].

For standard general relativity (GR), substantial progress has been made, and extensive waveform catalogs now cover much of the parameter space, particularly for near-equal-mass systems (We here define the mass ratio q as $q = \frac{M_{\text{larger}}}{M_{\text{smaller}}} \geq 1$). The LIGO–Virgo–KAGRA (LVK) collaboration has reported hundreds of detected events using these models [15]. However, two key challenges remain: improving the fidelity of GR simulations and developing waveform catalogs for alternative theories of gravity .

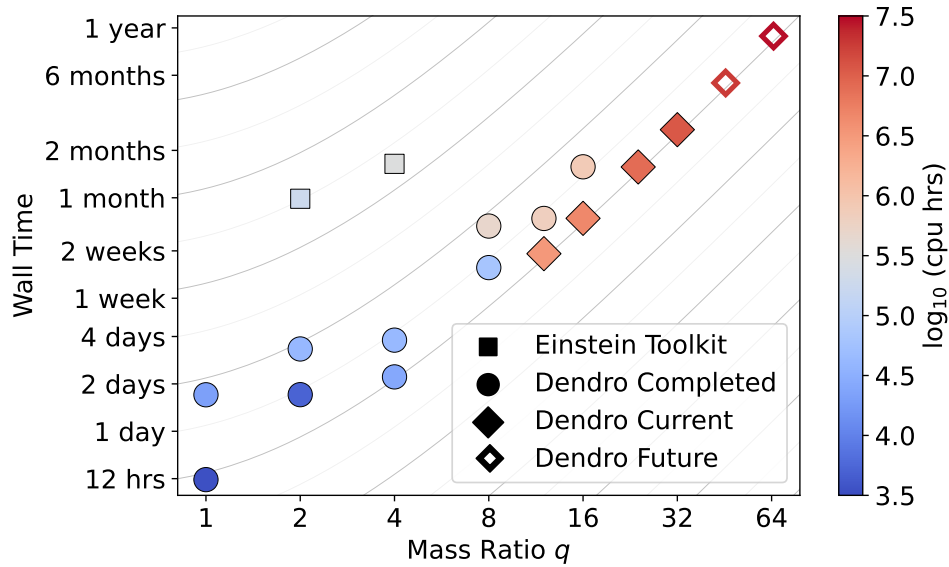


Figure 1.2 Walltime of numerical relativity simulations as a function of mass ratio.

The next generation of gravitational wave detectors is expected to significantly improve the signal-to-noise ratio of observed signals. Meeting this challenge will require higher-fidelity numerical simulations and more accurate waveform models.

In general, higher-fidelity simulations—particularly those with larger mass ratios—require finer grid resolution, leading to a substantial increase in computational cost. This trend is illustrated in Fig. 1.2. Current simulations with mass ratios $q < 8$ can require weeks of runtime on modern supercomputers, and further increases in resolution would significantly amplify this cost.

To obtain more accurate waveforms, there are two main approaches: running existing codes at higher resolution or improving the underlying numerical methods. A drawback of the brute-force approach—simply increasing resolution—is that the timescale of the problem is constrained by the CFL factor. This, in turn, requires more timesteps, significantly increasing the total runtime of the simulation.

One potential avenue for improving efficiency in numerical relativity codes lies in the computation of spatial derivatives. Most current implementations rely on explicit finite difference methods, while

spectral methods—although highly accurate—are often difficult to implement. In this work, we explore compact finite difference (CFD) operators as an alternative. Compact schemes are implicit methods that can achieve higher spectral resolution than comparable explicit schemes [16]. In tests within our Maxwell solver, CFDs often perform comparably to explicit methods at approximately half the resolution. Despite their success in computational fluid dynamics and related fields, their use in numerical relativity remains limited [17].

Gravitational wave observations provide a powerful probe of gravity in the strongly nonlinear, dynamical regime, where deviations from general relativity (GR) may become apparent. To search for such deviations, waveform templates from alternative theories of gravity are required. While some work has been carried out in theories such as Chern–Simons gravity, scalar–tensor theories, Einstein–Gauss–Bonnet gravity, and Einstein–Maxwell–dilaton theory, waveform catalogs in these frameworks remain incomplete.

In this dissertation, we focus on Einstein–Maxwell–dilaton–axion (EMDA) theory. EMDA is of particular interest because it admits a well-posed initial value formulation and arises as the low-energy limit of string theory. In contrast to the no-hair theorem in standard GR, which states the black holes can be completely defined by their mass spin and charge, EMDA black holes possess additional degrees of freedom associated with the dilaton and axion fields.

We present a numerical relativity framework for simulating binary black hole mergers in EMDA theory. We analyze the resulting waveforms and investigate whether deviations from GR may be observable in gravitational wave signals.

1.2 Background

We focus on the study of black hole dynamics. As such, we begin with Einstein’s field equations:

$$G_{ab} = 8\pi T_{ab} + \Lambda g_{ab}. \quad (1.1)$$

The left-hand side describes the geometry of spacetime, while the right-hand side encodes the matter and energy content. [18, 19] We briefly examine each term in turn.

The tensor G_{ab} is the Einstein tensor, which is constructed from quantities that encode the curvature of spacetime. The stress-energy tensor T_{ab} describes the distribution of matter and energy. The metric tensor g_{ab} defines distances and causal structure in spacetime. The constant Λ is the cosmological constant, which is generally assumed to be very small. On the scales considered in this work, it is negligible, and we therefore set $\Lambda = 0$.

In standard general relativity simulations of binary black hole mergers, we consider vacuum spacetimes, for which

$$G_{ab} = 0. \quad (1.2)$$

This provides the system of partial differential equations that must be solved in vacuum GR. Before proceeding further, we review some essential geometric concepts.

We begin with the metric, which encodes the notion of separation in spacetime. As an example, the Minkowski metric describing flat spacetime can be written as

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2, \quad (1.3)$$

or equivalently in matrix form as

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (1.4)$$

Here we use η_{ab} to denote the flat spacetime metric. The negative sign in the time component reflects our choice of signature $(-, +, +, +)$.

The Minkowski metric can also be expressed in different coordinate systems. For example, in spherical coordinates it takes the form

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}. \quad (1.5)$$

Although these representations appear different, they describe the same flat spacetime. This illustrates that the form of the metric depends on the choice of coordinates.

A natural question is how to determine whether a given metric describes flat spacetime. A necessary and sufficient condition is that the Riemann curvature tensor vanishes everywhere. Given a metric, the Riemann tensor is defined as

$$R^a{}_{bcd} = \partial_c \Gamma^a{}_{bd} - \partial_d \Gamma^a{}_{bc} + \Gamma^a{}_{ec} \Gamma^e{}_{bd} - \Gamma^a{}_{ed} \Gamma^e{}_{bc}, \quad (1.6)$$

where $\Gamma^a{}_{bc}$ are the Christoffel symbols, given by

$$\Gamma^a{}_{bc} = \frac{1}{2} g^{ad} (\partial_b g_{dc} + \partial_c g_{bd} - \partial_d g_{bc}). \quad (1.7)$$

These quantities encode how the coordinate system varies from point to point.

Contracting the Riemann tensor yields the Ricci tensor,

$$R_{ab} = g^{cd} R_{acbd}, \quad (1.8)$$

and further contraction gives the Ricci scalar,

$$R = g^{ab} R_{ab}. \quad (1.9)$$

The Einstein tensor is then constructed as

$$G_{ab} = R_{ab} - \frac{1}{2} g_{ab} R, \quad (1.10)$$

which satisfies the contracted Bianchi identities and ensures local conservation of energy and momentum.

With the Einstein tensor in hand, we can rewrite the field equations as a set of coupled second-order partial differential equations. To evolve these equations numerically, we must decompose them into a form suitable for time evolution. This requires splitting spacetime into three spatial dimensions and one time dimension.

The most straightforward approach leads to the ADM (Arnowitt–Deser–Misner) formulation. However, the ADM system is only weakly hyperbolic and can become unstable over long evolutions. Several modified formulations have been developed to address this issue. In this work, we use the BSSN (Baumgarte–Shapiro–Shibata–Nakamura) formulation, which is strongly hyperbolic and can be evolved stably for long periods of time. Given this formulation, we require suitable initial data, after which the system can be evolved forward in time.

To provide an overview of this procedure, we briefly consider simpler examples: a scalar field and Maxwell’s equations.

We begin with the scalar wave equation with a source term:

$$\square\psi = \nabla_a \nabla^a \psi = 4\pi\rho. \quad (1.11)$$

This equation is written in fully covariant form. Expanding it in flat spacetime using the Minkowski metric, we obtain

$$\square\psi = \eta^{ab} \nabla_a \nabla_b \psi = \left(-\partial_t^2 + D^2\right)\psi, \quad (1.12)$$

where D denotes purely spatial derivatives.

This provides a simple example of a 3+1 decomposition. The scalar wave equation is a second-order hyperbolic equation with well-posed initial value properties: solutions exist and are unique. Importantly, there are no constraint equations in this system, meaning we are free to choose ψ and ρ arbitrarily as initial data. As we will see, this is not the case for Maxwell’s equations or general relativity, where constraint equations must be satisfied.

We now turn to the more complex case of Maxwell's equations:

$$\nabla \cdot \mathbf{E} = 4\pi\rho, \quad (1.13)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (1.14)$$

$$\frac{\partial \mathbf{E}}{\partial t} = \nabla \times \mathbf{B} - 4\pi\mathbf{J}, \quad (1.15)$$

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E}. \quad (1.16)$$

We present the full 3 + 1 decomposition of Maxwell's equations in the appendix. However, an important feature is already evident: there are eight equations, but only six dynamical variables $(E_x, E_y, E_z, B_x, B_y, B_z)$. The remaining two equations act as constraints on the system.

Maxwell's equations can be separated into evolution equations:

$$\frac{\partial \mathbf{E}}{\partial t} = \nabla \times \mathbf{B} - 4\pi\mathbf{J}, \quad (1.17)$$

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E}, \quad (1.18)$$

and constraint equations:

$$\nabla \cdot \mathbf{E} = 4\pi\rho, \quad (1.19)$$

$$\nabla \cdot \mathbf{B} = 0. \quad (1.20)$$

Specifically, Gauss's law and the absence of magnetic monopoles must be satisfied at all times. If the initial data satisfy these constraints, then in the continuum limit the evolution equations will preserve them. However, in numerical simulations we work with finite precision, and violations of the constraint equations provide a useful measure of numerical error.

Solving the Cauchy problem for Einstein's equations is nontrivial, as we must satisfy both the Hamiltonian and momentum constraints. There are established techniques for solving these constraints. In particular, the momentum constraints are often addressed using the Bowen–York formalism. The Hamiltonian constraint reduces to an elliptic equation that must be solved numerically.

In EMDA, solving the Cauchy problem is further complicated by the presence of additional source terms. The stress-energy tensor T_{ab} encodes all contributions from matter and energy apart from gravity. In standard GR this is 0, in EMDA, this includes electromagnetic fields as well as scalar fields. The solution for single black holes is known analytically for the certain coupling strengths ($\alpha_0 = \alpha_1 = 1$) but numerically one could also imagine varying the coupling strength to see how the evolutions change. This dissertation will address the challenges associated with constructing consistent initial data and evolving these systems.

Once we have constructed suitable initial data and evolved the system, we obtain a gravitational waveform. We compute this waveform using the Newman–Penrose formalism, in which it is represented by the scalar Ψ_4 . Details of how Ψ_4 is calculated are provided in the appendix.

From Ψ_4 , we can reconstruct the gravitational-wave strain measured by LIGO. This allows us to compare our simulated waveforms with observed signals. To quantify the difference between two waveforms, we compute the mismatch, defined as

$$\text{mismatch}(h_1, h_2) = 1 - \max \mathcal{O}(h_1, h_2), \quad (1.21)$$

where the maximum is taken over time shifts and polarization angles.

From the mismatch, we can estimate the minimum signal-to-noise ratio (SNR) required to distinguish between the two waveforms. This is given by

$$\text{SNR}(h_1, h_2) = \frac{1}{\sqrt{2 \text{mismatch}(h_1, h_2)}}. \quad (1.22)$$

Despite this framework, interpreting differences between waveforms remains challenging. The parameter space is often degenerate; for example, it may not be immediately clear whether observed differences arise from spin, charge, or other physical effects.

We additionally present the work we have done incorporating compact finite differences into our numerical relativity code. To begin, we first discuss the methods currently used in the code.

On a discrete grid, we can approximate derivatives using finite differences. For example, a second-order accurate centered approximation to the first derivative is given by

$$\left. \frac{df}{dx} \right|_i = \frac{f_{i+1} - f_{i-1}}{2\Delta x} + O(\Delta x^2). \quad (1.23)$$

For higher-order accuracy, we can use a Taylor series expansion to construct more accurate approximations. For instance, a fourth-order accurate stencil is

$$\left. \frac{df}{dx} \right|_i = \frac{-f_{i+2} + 8f_{i+1} - 8f_{i-1} + f_{i-2}}{12\Delta x} + O(\Delta x^4). \quad (1.24)$$

These are examples of explicit finite difference schemes. Alternatively, we can seek an approximation to the first derivative using a compact (implicit) scheme of the form

$$\beta f'_{i-2} + \alpha f'_{i-1} + f'_i + \alpha f'_{i+1} + \beta f'_{i+2} = c \frac{f_{i+3} - f_{i-3}}{6\Delta x} + b \frac{f_{i+2} - f_{i-2}}{4\Delta x} + a \frac{f_{i+1} - f_{i-1}}{2\Delta x}. \quad (1.25)$$

The coefficients can be determined by matching Taylor series expansions to the desired order of accuracy (a more detailed derivation is provided in the appendix):

$$a + b + c = 1 + 2\alpha + 2\beta \quad (2\text{nd order}), \quad (1.26)$$

$$a + 2^2b + 3^2c = 2 \frac{3!}{2!} \left(\alpha + 2^2\beta \right) \quad (4\text{th order}), \quad (1.27)$$

$$a + 2^4b + 3^4c = 2 \frac{5!}{4!} \left(\alpha + 2^4\beta \right) \quad (6\text{th order}). \quad (1.28)$$

This approach is more complex and leads to an implicit scheme. However, the trade-off is improved accuracy and enhanced spectral resolution compared to explicit methods. This improvement in spectral resolution is illustrated in Fig. 1.3.

There are many possible compact finite difference operators that can be constructed. In this work, we describe our methodology for identifying accurate and stable compact derivative operators and evaluate their performance relative to their explicit counterparts.

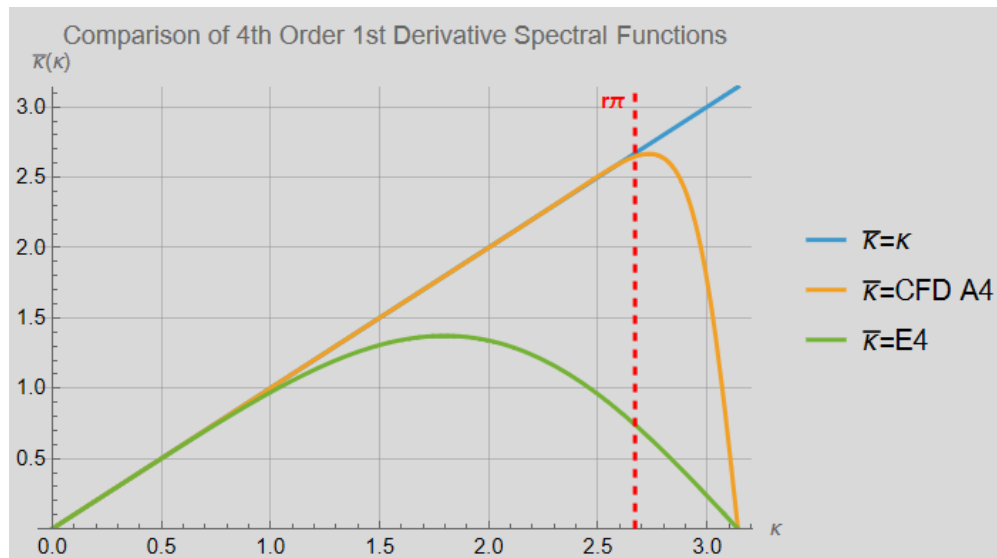


Figure 1.3 Comparison of spectral resolution for explicit and compact fourth-order first-derivative operators.

1.3 Summary of Dissertation

In this dissertation, we primarily present results from binary black hole simulations in EMDA theory. The dissertation is organized as follows. Chapter 2 discusses the formulation of the initial data and presents results from head-on collision simulations. Chapter 3 presents inspiral and merger results in EMDA theory. Chapter 4 explores interpolated EMDA models in which the coupling strengths are varied. Chapter 5 investigates the implementation of compact finite difference operators in numerical relativity codes.

The results presented in this dissertation are currently in preparation for publication.

Each of these projects involved multiple collaborators. The EMDA work builds on earlier contributions by Hyun Lim (LANL) and was carried out collaboratively with Eric Hirschmann (BYU), David van Komen (University of Utah), Sebastian Vander Ploeg Fallon (Cornell), and David Neilsen (BYU). The compact finite difference (CFD) project was conducted in collaboration with

Eric Hirschmann (BYU), David van Komen (University of Utah), David Neilsen (BYU), as well as Nathaneil Garey and Luke Papenfuss (BYU).

1.4 Conventions

In this work, we adopt geometrized units such that

- $c = G = 1$.
- Lengths and masses are expressed in units of a characteristic mass scale M .
- Distance: $d = M \Rightarrow d = GM/c^2$.
- Time: $t = M \Rightarrow t = GM/c^3$.
- We use the metric signature $(-, +, +, +)$.
- Einstein summation convention is assumed unless otherwise stated.
- Spacetime indices $a, b, c, \dots$ run over $0, 1, 2, 3$, while spatial indices $i, j, k, \dots$ run over $1, 2, 3$.
- The symmetric and antisymmetric parts of a tensor are defined as

$$T_{(ab)} \equiv \frac{1}{2} (T_{ab} + T_{ba}), \quad T_{[ab]} \equiv \frac{1}{2} (T_{ab} - T_{ba}).$$

- Partial derivatives are denoted by $\partial_i g$ or equivalently $g_{,i}$.
- Covariant derivatives are written as $\nabla_a f_b$ or equivalently $f_{b;a}$.
- The Lie derivative of a tensor X^a along a vector field β^a is denoted by $\mathcal{L}_\beta X^a$.
- In general, we refer to objects associated with

– the spacetime manifold $\mathcal{M}$ as g_{ab} , ${}^{(4)}\Gamma^a_{bc}$, ∇_a , ${}^{(4)}R_{ab}$, etc.,

- a spatial slice Σ as γ_{ij} , Γ^i_{jk} , D_i , R_{ij} , etc.,
- a conformally related space as $\bar{\gamma}_{ij}$, $\bar{\Gamma}^i_{jk}$, $\bar{D}_i$, $\bar{R}_{ij}$, etc.

Chapter 2

EMDA Chapter 1

2.1 Introduction

The increasing sensitivity of gravitational-wave detectors has enabled progressively more stringent tests of General Relativity (GR) [20–22]. Recent advances in gravitational-wave astronomy have led to the detection of signals from compact binary mergers, granting direct access to strongly gravitating, nonlinear regimes. All observed mergers to date are consistent with the predictions of GR [4, 15, 23].

Despite this agreement, the long-standing tension between GR and quantum mechanics remains unresolved. The broader field of quantum gravity seeks to address this discrepancy, motivating the exploration of alternative theories of gravity. Examples include Horndeski gravity [24], Lovelock gravity [25, 26], scalar Gauss–Bonnet gravity [27–29], and Einstein–Maxwell–Dilaton–Axion (EMDA) theory [30, 31].

In this work, we focus on the EMDA framework, examining the stability of its black-hole solutions as a prerequisite for producing inspiral waveforms.

As detector sensitivity improves and additional signals are recorded, increasingly precise tests of GR will become possible [32, 33]. Waveforms predicted by alternative theories play a dual role: they guide searches for new signatures and provide benchmarks for testing those theories. Such comparisons may ultimately reveal measurable deviations from GR. Alternative theories generally lack the extensive waveform catalogs available in GR; current efforts often rely on phenomenological approaches, including post-Newtonian expansions and numerical-relativity simulations [34, 35].

In standard GR, black holes are “bald”—that is, they are completely described by their mass, spin, and charge [36, 37]. By contrast, several extensions of GR predict scalarized black holes. For instance, recent studies have examined scalarization during mergers in scalar Gauss–Bonnet gravity [28, 29]. The EMDA theory, which arises in the low-energy limit of heterotic string theory, introduces both a scalar dilaton and an axion while retaining a well-posed initial-data formulation.

The existence of scalar hair in EMDA depends on charge and spin. In the limit of vanishing axion coupling, EMDA reduces to the simpler Einstein–Maxwell–Dilaton (EMD) theory.

Although numerical simulations of binary black holes in the EMDA theory remain limited, progress has been made in the Einstein–Maxwell–Dilaton setting [38, 39], establishing a foundation for extending such studies to the full EMDA framework. It is not immediately clear whether the scalar hair predicted by the EMDA theory remains stable or eventually radiates away, leaving behind a remnant black hole characterized solely by its mass, spin, and charge. Previous studies have explored the related phenomena of scalarization and descalarization in other modified gravity theories [40–42]. EMDA is a particularly compelling theory as it allows for direct tests of low energy string theory. Establishing the stability of Kerr–Sen black holes therefore represents an important direction for future investigation.

Previous work has explored astrophysical observations for potential evidence of Kerr–Sen black holes. Analyses of Event Horizon Telescope (EHT) shadow data suggest that Sgr A* favors the Kerr–Sen solution, while M87* favors Kerr but remains consistent with Kerr–Sen within current observational uncertainties [43, 44].

In this study, we extend these investigations by simulating head-on mergers governed by the full EMDA equations. We analyze the resulting waveforms and examine the stability of the remnants, thereby gaining insight into the persistence of scalar hair within the theory. Specifically, we evolve several head-on collisions of black holes in EMDA to determine the conditions under which the dilaton and axion fields survive the merger. An additional key question is to determine under what conditions initially unscalarized black holes will undergo scalarization.

2.2 Einstein–Maxwell–Dilaton–Axion

EMDA arises from low-energy string theory. This includes a U(1) gauge field and a scalar field that couples to the gauge field. In particular the action for low-energy heterotic string theory is::

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{8\pi G} - 2(\nabla\phi)^2 - e^{-2\alpha_0\phi} F^2 - \frac{1}{2} e^{4\alpha_1\phi} (\nabla\kappa)^2 - \kappa F_{ab} (*F)^{ab} \right]. \quad (2.1)$$

Here, g denotes the determinant of the metric, R the Ricci scalar, ϕ the dilaton, F_{ab} the Faraday tensor, and κ the axion. In the absence of charge, the equations reduce to the standard Einstein equations. In contrast to the no-hair theorem, the EMDA theory admits two additional fields—the dilaton ϕ and the axion κ —which couple to the electromagnetic field and together constitute the “hair.” The parameters α_0 and α_1 set the coupling strengths; in this work we take $\alpha_0 = \alpha_1 = 1$, corresponding to the low-energy limit of heterotic string theory. Setting $\alpha_0 = \alpha_1 = 0$ reduces the theory to Einstein–Maxwell with minimally coupled scalar and axion fields.

To perform numerical evolutions of Kerr–Sen black holes, we formulate the equations as a Cauchy problem and use the BSSN formulation [45–47] of the Einstein equations, supplemented with the necessary terms to account for electromagnetic fields, axion and dilaton. We also employ the puncture gauge condition [48, 49]. Initial data are prepared for head-on black hole collisions. The full equations of motion and the 3 + 1 decomposition are presented in Appendix H.1.

2.3 Numerical methods

2.3.1 Code Description

For these simulations, we modified the existing numerical-relativity framework Dendro-GR. Dendro-GR is a portable code that combines an octree-based adaptive multiresolution framework with automatic code generation for solving the EMDA equations. Spatial derivatives are

computed using sixth-order finite-difference stencils, and time integration is performed with a fourth-order Runge–Kutta scheme. More details on Dendro–GR can be found in [12, 50–52].

2.3.2 Initial Data

Valid initial data must satisfy the constraint equations; in particular, we must satisfy the Hamiltonian, momentum, and Maxwell constraints. In natural units, these are given by

$$R + K^2 - K_{ij}K^{ij} = 16\pi\rho, \quad (2.2)$$

$$D_j(K^{ij} - \gamma^{ij}K) = 8\pi J^i, \quad (2.3)$$

where $\mathcal{R}$ is the three-dimensional Ricci scalar associated with the metric, K_{ij} is the extrinsic curvature, K is the trace of the extrinsic curvature, ρ is the energy density, and J^i is the current density.

$$D_i E^i = 0, \quad D_i B^i = 0, \quad (2.4)$$

The energy and current densities are defined as

$$\begin{aligned} \rho \equiv & \Pi^2 + (D\phi)^2 + e^{-2\alpha_0\phi} (E_a E^a + B_a B^a) \\ & + \frac{1}{4} e^{4\alpha_1\phi} (\Xi^2 + (D\kappa)^2), \end{aligned} \quad (2.5)$$

$$J^i \equiv 2\Pi D^i \phi + 2e^{-2\alpha_0\phi} \epsilon^{ijk} E_j B_k + \frac{1}{2} e^{4\alpha_1\phi} \Xi D^i \kappa. \quad (2.6)$$

The EMDA theory possesses a mathematical structure that admits well-posed initial data. The formulation of such data has not been extensively investigated in the literature, although there have been some approaches to generating initial data for charged black holes [53–57]. In this work, we adapt techniques originally developed for constructing the initial data of charged black holes. Specifically, we employ an elliptic solver based on a modified version of the two-charged-punctures solver introduced in [58], augmented to incorporate contributions from the dilaton. A more detailed discussion of the initial-data construction is provided in Appendix H.2.

To analyze the numerical data, we compute the Newman–Penrose scalar Ψ_4 , as well as the radiated electromagnetic quantities, extracting their multipole moments at $r \approx 100M$. Further details on the Waveform extraction are in Appendix C.

2.3.3 Simulation Setup

A summary of the black hole parameters used in the simulations is provided in Appendix H.4.

2.4 Results

We performed tests on a number of black holes in the nonextremal limit of the parameter space; our search was limited to equal–mass–ratio black holes with low to moderate charge and low spin. We present here a representative example of the waveform and energy radiated. The total mass of the system is 1, and the charge of the system lies between $0.0625 \leq \frac{Q}{M} \leq 0.5$, with spins in the range $0.05 \leq \chi \leq 0.2$. A full summary can be seen in Appendix C. The waveforms of a head-on collision corresponding to two identical black holes with $\chi = 0.2$ and $\frac{Q}{M} = 0.5$ at an initial separation of $x = 8$ are shown in Fig. ???. The results from this configuration are largely representative of the overall behavior observed in our tests.

The radiated energy as a function of time is shown in Fig. 2.1.

In this case, the gravitational wave is the dominant form of energy released. The electromagnetic radiation is the next most significant, followed by the dilaton radiation. The contribution from the axion remains comparatively small. We see that the energy from the scalar fields quickly dies off after the merger occurs. This indicates that once the merger has taken place, the field remains in a relatively stable state. In our experiments, we find that the scalar fields persist and do not radiate away after the merger.

A non-trivial axion field persists after the merger, as illustrated in Fig. ??.

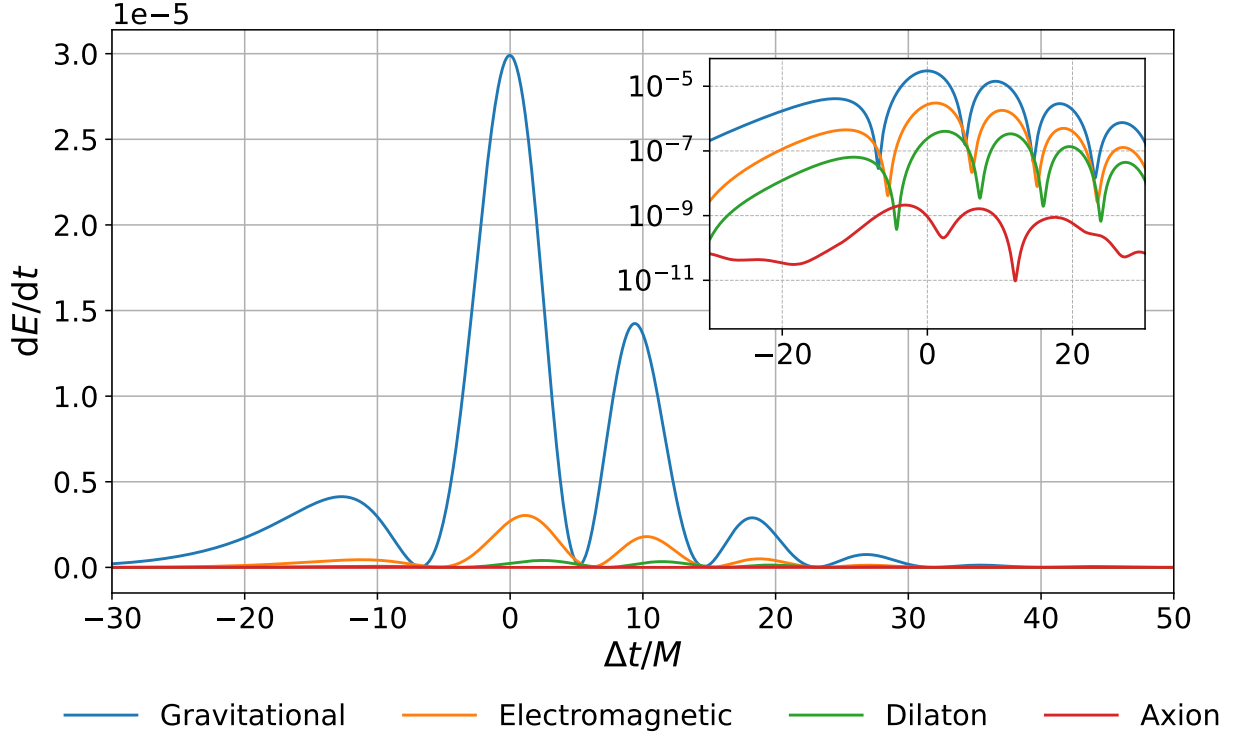


Figure 2.1 Radiated fluxes for the case $\chi = 0.2$ and $Q/M = 0.5$. The curves have been aligned such that $t = 0$ corresponds to the global maximum. The inset plots show the same data with the y-axis on a logarithmic scale.

2.4.1 Initially Unscalarized Black Holes

To further assess the stability of scalar hair, we conducted a series of tests initialized with a Gaussian distribution of axion and dilaton fields centered at the origin, with amplitudes ranging from 10^{-1} to 10^{-7} . These initial perturbations propagate outward, enabling us to examine whether initially unscalarized black holes can acquire and sustain scalar fields. Our results confirm that they do: a Kerr–Newman black hole in the EMDA theory rapidly scalarizes, developing both dilaton and axion hair.

Remarkably, even in the absence of any initial scalar fields in the background, the system evolves to generate nontrivial scalar structure. This demonstrates that scalarization occurs dynamically and does not require preexisting scalar hair. The conserved quantities, including the total mass and

charge, remain effectively constant throughout the evolution. Variations in the conserved quantities over the course of the evolution in both mass and charge are below 0.2%. This indicates that our quantities remain stable and conserved overtime. Figure ?? illustrates the evolution of the scalar fields for the case without initial scalar hair.

2.5 Conclusion

We have simulated a range of different parameters for the head-on collisions of Kerr–Sen black holes. We have primarily been concerned with the persistence of hair. In summary, our simulations show that the scalar fields do not radiate away entirely, so long as there is a nonzero charge (and spin for the axion). We have found that Kerr–Sen black holes are stable. In particular, we find that if the remnant black hole has a nonzero charge, the dilaton field persists. If it has both nonzero charge and spin, the axion field also persists after merger. These results suggest that Kerr–Sen black holes are stable under the conditions and time frames we considered.

A natural follow-up study would be the analysis of waveforms produced by inspiraling black holes, an area where the available results remain relatively limited. In this paper, we have further assumed that we are working within the low-energy sector of string theory. A more detailed investigation of the theory space parameterized by α_0 could be intriguing; in particular, the coupling $\alpha_0 = \sqrt{3}$ corresponds to the Kaluza–Klein theory [59]. Future work may explore interpolations between these theories.

Chapter 3

EMDA Chapter 2

3.1 Introduction

Recent improvements in gravitational-wave detectors have opened a new regime for precision tests of general relativity (GR). Current observations already reach signal-to-noise ratios high enough to place increasingly stringent constraints on deviations from GR [?]. As detector sensitivity continues to improve, gravitational waves will provide an increasingly powerful probe of strong-field gravity, where compact objects probe highly dynamical and nonlinear spacetime curvature.

This motivates the study of well-posed alternative theories of gravity and their predictions for compact-binary dynamics. A number of theories beyond GR have been considered in this context, including scalar-tensor theories, dynamical Chern–Simons gravity, and Einstein–Gauss–Bonnet gravity. In several of these theories, black holes can acquire additional structure or “hair,” leading to modified inspiral dynamics, merger behavior, or gravitational-wave emission. However, despite this progress, comparatively less work has been done on fully dynamical binary black hole evolutions in Einstein–Maxwell–dilaton–axion (EMDA) theory.

EMDA theory is a particularly interesting target for numerical study for several reasons. It provides a well-posed extension of GR coupled to electromagnetic, dilaton, and axion fields, and it arises as a low-energy limit of heterotic string theory. The theory also admits rotating charged black-hole solutions, such as the Kerr–Sen spacetime, which differ from Kerr black holes while remaining observationally viable in some regimes. Previous work has investigated the stability of black-hole hair in EMDA theory [60], and black-hole shadow studies using Event Horizon Telescope observations have not ruled out Kerr–Sen black holes.

These facts motivate a systematic study of binary black hole dynamics in EMDA theory. In this work, we investigate binary inspirals in EMDA and examine how the additional fields modify the dynamics and gravitational-wave signal relative to GR. Our goal is to provide a step toward connecting EMDA theory with future strong-field gravitational-wave tests.

The cases we consider here are more highly charged than are astrophysically likely. There have been a few mechanisms proposed by which black holes may obtain a charge but any charges are likely to be quite small. We have previously established the stability of EMDA black holes.

In this work, we present a modest catalog of binary black-hole waveforms in Einstein–Maxwell–dilaton–axion (EMDA) theory, considering several choices of mass ratio, spin, and charge. The parameter space of compact-binary waveforms is highly degenerate. Consequently, it is not always clear whether deviations from a GR waveform arise from genuinely new physics or whether they could instead be mimicked by changes in the intrinsic binary parameters, such as the mass ratio, spins, charges, or total mass scale.

A complete study of these degeneracies would require a full parameter-estimation analysis over the EMDA and GR parameter spaces, which is beyond the scope of this work. Instead, we consider a more restricted question. We assume that the spins are independently constrained and compare EMDA waveforms with corresponding GR waveforms while allowing for a rescaling of the total mass. This comparison allows us to test whether the dominant differences between EMDA and GR waveforms can be absorbed into a simple mass rescaling, or whether theory-dependent residuals remain.

3.1.1 **Goal of the Work**

The primary goal of this work is to determine under what circumstances EMDA waveforms differ noticeably from their GR counterparts. We focus on how the additional EMDA fields and charges modify the inspiral dynamics and gravitational-wave signal. By comparing EMDA and GR waveforms after accounting for possible mass rescalings, we identify cases in which the waveform differences remain distinguishable.

The waveform catalog presented here is intended as an initial step toward a more complete study of EMDA binary black-hole phenomenology. While we do not attempt to cover the full parameter

space or perform a complete parameter-estimation analysis, these results provide useful guidance for future studies aimed at determining whether high-precision gravitational-wave observations can constrain or detect signatures of EMDA theory.

3.1.2 Structure of the Paper and Conventions

The remainder of this paper is organized as follows. In Sec. 3.2, we review the EMDA equations, describe the numerical methods used for the evolutions, and discuss the construction of initial data. We then describe the procedure used to reduce orbital eccentricity and the waveform-extraction methods used in the analysis. In Sec. 3.4, we compare EMDA and GR waveforms and identify the circumstances under which their differences may become noticeable. We summarize our conclusions and discuss future directions in Sec. 3.5.

Unless otherwise stated, we use geometrized units with $G = c = 1$. Greek indices denote spacetime components, while Latin indices denote spatial components.

3.2 Methods

3.3 Einstein–Maxwell–Dilaton–Axion Theory

The action for EMDA theory is

$$S = \int d^4x \sqrt{-g} \left[R - 2(\nabla\phi)^2 - e^{-2\alpha_0\phi} F^2 - \frac{1}{2} e^{4\alpha_1\phi} (\nabla\kappa)^2 - \kappa F_{ab} (*F)^{ab} \right]. \quad (3.1)$$

Here, g denotes the determinant of the spacetime metric, R is the Ricci scalar, ϕ is the dilaton field, F_{ab} is the Faraday tensor, and κ is the axion field. The parameters α_0 and α_1 control the coupling strengths of the dilaton and axion sectors. In this work, we take $\alpha_0 = \alpha_1 = 1$, corresponding

to the low-energy limit of heterotic string theory. Setting $\alpha_0 = \alpha_1 = 0$ reduces the theory to Einstein–Maxwell theory with minimally coupled scalar and axion fields.

In the absence of charge, the electromagnetic field vanishes and the equations reduce to the standard Einstein equations. When charge is present, however, the electromagnetic field sources the dilaton and axion fields. In this sense, EMDA black holes possess additional field structure beyond the Kerr family. These extra fields provide a possible source of deviations from GR in compact-binary dynamics and gravitational-wave emission.

To perform numerical evolutions of Kerr–Sen black holes, we formulate the equations as a Cauchy problem and use the BSSN formulation [45–47] of the Einstein equations, supplemented by the electromagnetic, dilaton, and axion source terms. We also employ the moving-puncture gauge conditions [48, 49]. The full equations of motion and their 3 + 1 decomposition are presented in our earlier paper.

3.3.1 Controlling the Eccentricity

We use the same TPID solver described in our earlier work to construct binary black-hole initial data in EMDA theory. To set approximately quasicircular initial data, we use a post-Newtonian prescription implemented in the script developed by Zachariah Etienne. In applying this procedure, we assume that the direct contributions from the dilaton and axion fields to the initial orbital parameters are negligible.

We then follow the procedure used for quasicircular charged black-hole initial data in Ref. [61]. Although this approach is approximate, it produces relatively low eccentricity for the spinless configurations considered here. After excluding the first orbit, we find eccentricities of approximately **[insert eccentricities here]**.

3.3.2 Wave Extraction

To analyze the numerical data, we compute the Newman–Penrose scalar Ψ_4 and extract its multipole moments at finite radius. We also compute the corresponding radiated electromagnetic quantities. Unless otherwise stated, waveform quantities are extracted at $r \approx 100M$. Further details of the waveform-extraction procedure are given in Appendix ??.

3.3.3 Code Description

For these simulations, we modified the existing numerical-relativity framework Dendro-GR. Dendro-GR is a portable code that combines an octree-based adaptive multiresolution framework with automatic code generation for solving relativistic field equations. In this work, we extend the code to evolve the EMDA equations.

Spatial derivatives are computed using sixth-order finite-difference stencils, and time integration is performed using a fourth-order Runge–Kutta scheme. Additional details on Dendro-GR and its numerical infrastructure can be found in Refs. [12, 50–52].

3.3.4 Simulation Setup

A summary of the black-hole parameters used in the simulations is provided in Appendix ?. The catalog includes configurations with different mass ratios, spins, and charges. These simulations are designed to sample representative regions of the EMDA parameter space rather than to provide a complete parameter survey.

3.4 Results

We compare EMDA and GR waveforms for a number of representative configurations. In particular, we examine how the waveform differences depend on charge, mass ratio, and spin. We also compute

signal-to-noise ratios and compare strains after allowing for an overall mass rescaling. This allows us to assess whether EMDA waveforms can be effectively mimicked by nearby GR configurations or whether residual differences remain.

Our results suggest that, for astrophysically plausible charges, distinguishing EMDA waveforms from GR waveforms is likely to be challenging with current detector sensitivities. Larger charges can produce more noticeable deviations, but such configurations may be less astrophysically likely.

3.4.1 SNR and Strain Comparison

3.4.2 Detectability

3.5 Conclusion

In this work, we presented a modest catalog of binary black-hole waveforms in EMDA theory. We compared these waveforms with corresponding GR waveforms and examined whether the leading differences could be absorbed by a rescaling of the total mass. This provides an initial test of whether EMDA corrections produce distinguishable signatures in gravitational-wave observations.

We find that waveform differences are generally small for astrophysically plausible charges, suggesting that EMDA effects may be difficult to distinguish from GR in such regimes. However, for larger charges, the deviations become more pronounced and may provide useful targets for future studies. A more complete parameter-estimation analysis will be needed to determine whether these effects can be robustly distinguished from changes in mass ratio, spin, or other binary parameters.

Future work should extend this catalog to a broader region of the EMDA parameter space, include longer inspirals and higher-resolution simulations, and perform direct comparisons with detector noise curves and Bayesian parameter-estimation pipelines.

.1 Waveform Catalog

Appendix A

EMDA Chapter 3

Appendix B

CFD Chapter 1

B.1 CFD

Appendix C

Conclusion

C.1 Conclusion

Appendix A

Mathematical Overview

A.1 Electricity and Magnetism

We give here an example of how to perform a 3 + 1 decomposition for electricity and magnetism.

We follow an approach similar to that of *Numerical Relativity from Scratch* [14].

We begin with Maxwell's equations written in terms of the Faraday tensor:

$$\nabla_b F^{ab} = 4\pi j^a, \quad (\text{A.1})$$

and

$$\nabla_{[a} F_{bc]} = 0, \quad (\text{A.2})$$

where we have defined the four-current as

$$j^a = (\rho, j^i). \quad (\text{A.3})$$

We note that the continuity equation can be written as

$$\nabla_a j^a = 0. \quad (\text{A.4})$$

In general relativity, there is a similar conservation law associated with the Bianchi identity.

We further note that the Maxwell tensor can be written in terms of the vector potential A^a as

$$F^{ab} = \nabla^a A^b - \nabla^b A^a. \quad (\text{A.5})$$

With this definition, the second of Maxwell's equations is satisfied automatically. Substituting this expression into the first Maxwell equation, Eq. (A.1), gives

$$\nabla_b \nabla^a A^b - \nabla_b \nabla^b A^a = 4\pi j^a. \quad (\text{A.6})$$

Here A^a is the vector potential. In a manner similar to the four-current, we write the vector potential as

$$A^a = (\phi, A^i). \quad (\text{A.7})$$

We now first consider the time component of Eq. (A.6). Setting the index a to the time component, t , we have

$$\nabla_b \nabla^t A^b - \nabla_b \nabla^b A^t = \nabla_b \nabla^t A^b - \nabla_j \nabla^j \phi = 4\pi j^t = 4\pi \rho. \quad (\text{A.8})$$

Here we have changed the index from b to j because the remaining components are purely spatial. We also note that, when $b = t$, the two terms cancel. In special relativity, corresponding to flat spacetime, we make the following identifications:

$$\nabla_j = D_j, \quad \nabla_t = \partial_t. \quad (\text{A.9})$$

We also have $\nabla^t = \eta^{ta} \nabla_a = -\partial_t$, and therefore

$$D_j (\partial_t A^j - D^j \phi) = 4\pi \rho. \quad (\text{A.10})$$

We can now make the identification

$$E^i = F^{ti} = -\partial_t A^i - D^i \phi. \quad (\text{A.11})$$

This can also be rearranged to obtain the evolution equation

$$\partial_t A^i = -E^i - D^i \phi. \quad (\text{A.12})$$

As a result, we recover Gauss's law,

$$D_i E^i = 4\pi \rho, \quad (\text{A.13})$$

which gives the constraint equation.

We now consider the spatial components of Eq. (A.6), corresponding to $a = i$:

$$\nabla_t (\nabla^i A^t - \nabla^t A^i) = -\nabla_j \nabla^i A^j + \nabla_j \nabla^j A^i + 4\pi j^i. \quad (\text{A.14})$$

Here we have grouped the terms with $b = t$ on the left-hand side and the terms with $b = j$ on the right-hand side. We then obtain

$$\partial_t E_i = -D_j D^j A_i + D_i D^j A_j - 4\pi j_i. \quad (\text{A.15})$$

We now have both the evolution and constraint equations. We will follow the same pattern as in general relativity, but we will have to contend with non-flat spacetime.

To summarize, we have six evolution equations:

$$\partial_t E_i = -D_j D^j A_i + D_i D^j A_j - 4\pi j_i. \quad (\text{A.16})$$

$$\partial_t A^i = -E^i - D^i \phi. \quad (\text{A.17})$$

and two constraint equations:

$$D_i E^i = 4\pi \rho, \quad (\text{A.18})$$

$$D_i B^i = 0. \quad (\text{A.19})$$

A.2 Spacetime and Kinematics

We now allow for more general slices of spacetime. This increased freedom is needed for solving the Cauchy problem in general relativity and in alternative theories of gravity. The surfaces we embed in spacetime can be thought of as hypersurfaces. We assume that these hypersurfaces are spacelike, so that they represent slices of constant time.

We can define the gradient of t as

$$\nabla_a t = \partial_a t. \quad (\text{A.20})$$

This quantity is perpendicular to the constant-time hypersurfaces. We then introduce the normal n_a to the slices as proportional to this gradient:

$$n_a = -\alpha \nabla_a t. \quad (\text{A.21})$$

We choose n_a to have unit magnitude:

$$n_a n^a = g^{ab} n_a n_b = -1. \quad (\text{A.22})$$

Because the slices are spacelike, the normal four-vector must be timelike, and we therefore normalize it to negative unity. The quantity α measures the ratio of coordinate time to proper time. In general, α is a coordinate-dependent quantity.

In this coordinate system,

$$\nabla_a t = \partial_a t = \frac{\partial t}{\partial x^a}, \quad (\text{A.23})$$

so the only nonzero component occurs when $a = t$. The normal vector then reduces to

$$n_a = (-\alpha, 0, 0, 0). \quad (\text{A.24})$$

We can write α in terms of the proper time interval $d\tau$ as

$$d\tau = \alpha dt. \quad (\text{A.25})$$

We often call α the lapse function.

The normalization condition can be satisfied by writing

$$n^a = \frac{1}{\alpha}(1, \beta^i). \quad (\text{A.26})$$

This holds with our normalization condition for any β^i . Viewing n^a as the four-velocity of a normal observer, β^i measures the extent to which the spatial coordinates change as the normal observer moves through spacetime. This quantity is called the shift vector.

We can also use the normal vector to decompose the metric into spatial and temporal parts. We write

$$g^a_b = \gamma^a_b - n^a n_b. \quad (\text{A.27})$$

Here γ^a_b is the spatial part of the metric and is called the projection operator.

Changes in the normal vector determine how the spatial hypersurfaces are embedded in spacetime. We can decompose this as

$$\begin{aligned}
 \nabla_a n_b &= g_a^c g_b^d \nabla_c n_d \\
 &= (\gamma_a^c - n^c n_a) (\gamma_b^d - n^d n_b) \nabla_c n_d \\
 &= \gamma_a^c \gamma_b^d \nabla_c n_d - \gamma_a^c n_b n^d \nabla_c n_d - n_a n^c \gamma_b^d \nabla_c n_d + n_a n^c n_b n^d \nabla_c n_d.
 \end{aligned} \tag{A.28}$$

The Lie derivative measures how a tensor field changes as it is dragged along the flow generated by a vector field. For a covector A_d , the Lie derivative along n^a is

$$\mathcal{L}_n A_d = n^c \nabla_c A_d + A_c \nabla_d n^c. \tag{A.29}$$

Unlike the covariant derivative, the Lie derivative does not require any additional connection structure; it depends only on the vector field that generates the flow.

One important property is that the Lie derivative of a spatial tensor along αn^a is again spatial. Since we want our spatial quantities to remain spatial during the evolution, we decompose the coordinate time-evolution vector t^a into normal and tangential parts:

$$t^a = \alpha n^a + \beta^a, \tag{A.30}$$

or equivalently,

$$\alpha n^a = t^a - \beta^a. \tag{A.31}$$

Here α is the lapse function, which controls the rate of proper time elapsed between neighboring slices, and β^a is the shift vector, which describes how spatial coordinates move from one slice to the next.

We also define the extrinsic curvature by

$$K_{ab} = -\gamma_a^c \gamma_b^d \nabla_c n_d. \tag{A.32}$$

The minus sign is a convention that we adopt. Before we move on, we will also define the covariant derivative and the Christoffel symbols.

For the covariant derivative, we need a derivative operator that transforms like a tensor. For scalars, we can use the ordinary partial derivative, but for higher-rank tensors this is not sufficient. The partial derivative only accounts for changes in the components of the tensor, while ignoring changes in the basis vectors. To account for changes in the basis vectors, we introduce the Christoffel symbols, which are not themselves tensors.

For a vector A^b , the covariant derivative is defined by

$$\nabla_a A^b = \partial_a A^b + \Gamma^b_{ac} A^c. \quad (\text{A.33})$$

Here Γ^b_{ac} is the Christoffel symbol. For the Levi-Civita connection, which is compatible with the metric and torsion-free, the Christoffel symbols are given by

$$\Gamma^a_{bc} = \frac{1}{2} g^{ad} (\partial_b g_{dc} + \partial_c g_{db} - \partial_d g_{bc}). \quad (\text{A.34})$$

A.3 General Relativity

To decompose Einstein's equations in a similar way, we need the spacetime metric g_{ab} . Recall Einstein's equation:

$$G_{ab} + \Lambda g_{ab} = 8\pi T_{ab}. \quad (\text{A.35})$$

At first, this appears to provide ten equations for the ten independent components of G_{ab} . However, the Bianchi identity,

$$\nabla_a G^{ab} = 0, \quad (\text{A.36})$$

must always be satisfied. This plays a role similar to the constraints in electromagnetism. It also tells us that four of Einstein's equations are not independent. We therefore expect four constraint equations in general relativity, along with a fourfold gauge freedom.

We can now decompose both the variables and the equations. As mentioned above, the spacetime metric can be decomposed using the spatial projection operator:

$$g_{ab} = g^c{}_a g^d{}_b g_{cd} = (\gamma^c{}_a - n^c n_a) (\gamma^d{}_b - n^d n_b) g_{cd} = \gamma^c{}_a \gamma^d{}_b g_{cd} - n_a n_b. \quad (\text{A.37})$$

The first term on the right-hand side is

$$\gamma^c{}_a \gamma^d{}_b g_{cd} = g_{ab} + n_a n_b. \quad (\text{A.38})$$

This defines the spatial metric, which also acts as the projection tensor:

$$\gamma_{ab} = g_{ab} + n_a n_b. \quad (\text{A.39})$$

The spatial metric performs the same role on each spatial slice that the spacetime metric performs in the full four-dimensional spacetime. However, because γ_{ab} is purely spatial, its components normal to the slice vanish.

We can therefore write the inverse spacetime metric as

$$g^{ab} = \gamma^{ab} - n^a n^b = \begin{pmatrix} -\alpha^{-2} & \alpha^{-2} \beta^i \\ \alpha^{-2} \beta^j & \gamma^{ij} - \alpha^{-2} \beta^i \beta^j \end{pmatrix}. \quad (\text{A.40})$$

The line element then becomes

$$ds^2 = g_{ab} dx^a dx^b = -\alpha^2 dt^2 + \gamma_{ij} (dx^i + \beta^i dt) (dx^j + \beta^j dt). \quad (\text{A.41})$$

The quantity ds^2 encodes the notion of distance in spacetime and can be thought of as the spacetime analogue of the Pythagorean theorem.

The extrinsic curvature can also be related to the time evolution of the spatial metric. In particular, one can show that

$$K_{ab} = -\frac{1}{2} \mathcal{L}_n \gamma_{ab}. \quad (\text{A.42})$$

Using Eq. (A.31) and Eq. (A.42), this can be rewritten in terms of the coordinate time derivative as

$$\partial_t \gamma_{ij} = -2\alpha K_{ij} + \mathcal{L}_\beta \gamma_{ij}. \quad (\text{A.43})$$

Thus, the extrinsic curvature measures how the spatial metric changes in time, up to the lapse and shift. The spatial metric γ_{ij} and the extrinsic curvature K_{ij} together form the basic dynamical variables in the ADM decomposition.

From here, we need to further decompose the equations into purely spatial quantities. The spatial covariant derivative is defined in the same way as the spacetime covariant derivative, but with the connection associated with the spatial metric γ_{ij} . For a spatial vector V^j , we have

$$D_i V^j = \partial_i V^j + \Gamma^j_{ik} V^k. \quad (\text{A.44})$$

Here Γ^i_{jk} is the Christoffel symbol associated with the spatial metric:

$$\Gamma^i_{jk} = \frac{1}{2} \gamma^{il} (\partial_j \gamma_{lk} + \partial_k \gamma_{lj} - \partial_l \gamma_{jk}). \quad (\text{A.45})$$

To obtain the spatial decompositions of the Riemann and Ricci tensors, we use the Gauss–Codazzi relations. One useful projection is The spatial projections of the spacetime curvature are related to the intrinsic curvature of the spatial slice and the extrinsic curvature through the Gauss–Codazzi relations. These equations allow us to rewrite spacetime curvature quantities in terms of purely spatial quantities and the extrinsic curvature.

The Gauss equation is

$$\gamma^a_i \gamma^b_j \gamma^c_k \gamma^d_l {}^{(4)}R_{abcd} = {}^{(3)}R_{ijkl} + K_{ik} K_{jl} - K_{il} K_{jk}. \quad (\text{A.46})$$

Here ${}^{(4)}R_{abcd}$ is the four-dimensional spacetime Riemann tensor, while ${}^{(3)}R_{ijkl}$ is the three-dimensional Riemann tensor associated with the spatial metric γ_{ij} .

The Codazzi equation is

$$\gamma^a_i \gamma^b_j \gamma^c_k n^d {}^{(4)}R_{abcd} = D_j K_{ik} - D_i K_{jk}. \quad (\text{A.47})$$

This relates the mixed spatial-normal projection of the spacetime curvature to spatial derivatives of the extrinsic curvature.

The Ricci equation, which involves two projections along the normal direction, is

$$\gamma^a{}_i \gamma^b{}_j n^c n^{d(4)} R_{acbd} = \mathcal{L}_n K_{ij} + \frac{1}{\alpha} D_i D_j \alpha + K_{ik} K^k{}_j. \quad (\text{A.48})$$

Together, these relations provide the geometric bridge between the four-dimensional spacetime curvature and the three-dimensional geometry of each spatial slice. The preceding definitions come from differential geometry. To obtain the ADM equations, we now insert them into Einstein's equation and equate the projections of the Einstein tensor with the corresponding projections of the stress-energy tensor in Eq. (H.5).

Projecting Einstein's equation twice along the normal direction gives the Hamiltonian constraint:

$$R + K^2 - K_{ij} K^{ij} = 16\pi\rho. \quad (\text{A.49})$$

Here

$$\rho = n^a n^b T_{ab} \quad (\text{A.50})$$

is the energy density measured by an observer moving normal to the spatial slice. The quantity K is the trace of the extrinsic curvature,

$$K = \gamma^{ij} K_{ij}. \quad (\text{A.51})$$

A mixed projection, with one index projected along the normal direction and one projected onto the spatial slice, gives the momentum constraint:

$$D_j (K^{ij} - \gamma^{ij} K) = 8\pi j^i. \quad (\text{A.52})$$

Here

$$j^i = -\gamma^{ia} n^b T_{ab} \quad (\text{A.53})$$

is the momentum density measured by the normal observer.

A complete spatial projection gives the evolution equation for the extrinsic curvature:

$$\partial_t K_{ij} = \alpha \left(R_{ij} - 2K_{ik} K^k{}_j + K K_{ij} \right) - D_i D_j \alpha - 4\pi\alpha M_{ij} + \mathcal{L}_\beta K_{ij}. \quad (\text{A.54})$$

In this expression, we have used the abbreviation

$$M_{ij} = 2S_{ij} - \gamma_{ij} (S - \rho), \quad (\text{A.55})$$

where

$$S_{ij} = \gamma^a{}_i \gamma^b{}_j T_{ab} \quad (\text{A.56})$$

is the spatial projection of the stress-energy tensor, and

$$S = \gamma^{ij} S_{ij} \quad (\text{A.57})$$

is its trace.

Together with the evolution equation for the spatial metric,

$$\partial_t \gamma_{ij} = -2\alpha K_{ij} + \mathcal{L}_\beta \gamma_{ij}, \quad (\text{A.58})$$

these equations form the ADM system. The variables γ_{ij} and K_{ij} are the dynamical variables, while α and β^i encode the gauge freedom associated with the choice of spacetime coordinates.

$R + K^2 - K_{ij} K^{ij} = 16\pi\rho,$	Hamiltonian constraint,	(A.59)
$D_j (K^{ij} - \gamma^{ij} K) = 8\pi j^i,$	Momentum constraint,	
$\partial_t \gamma_{ij} = -2\alpha K_{ij} + \mathcal{L}_\beta \gamma_{ij},$	Metric evolution,	
$\partial_t K_{ij} = \alpha \left(R_{ij} - 2K_{ik} K^k{}_j + K K_{ij} \right) - D_i D_j \alpha$		
$- 4\pi\alpha [2S_{ij} - \gamma_{ij}(S - \rho)] + \mathcal{L}_\beta K_{ij},$	Extrinsic-curvature evolution.	

A.3.1 Reformulating the Einstein Equations

The ADM equations are actually only weakly hyperbolic, unfortunately they in general have been shown to be numerically unstable. We therefore need to Reformulate them. It has been shown how to reformaulte them into a strongly hyperbolic system of equations.

Reformulating Maxwell's equations

We begin by using the electromagnetism analogy. In a 3 + 1 decomposition, Maxwell's equations can be written as

$$\partial_t E_i = -D_j D^j A_i + D_i D^j A_j - 4\pi j_i, \quad (\text{A.60})$$

together with

$$\partial_t A_i = -E_i - D_i \phi. \quad (\text{A.61})$$

These evolution equations are supplemented by the constraint equation

$$D_i E^i = 4\pi \rho. \quad (\text{A.62})$$

We define the constraint violation C by

$$C \equiv D_i E^i - 4\pi \rho. \quad (\text{A.63})$$

Ideally, the constraint is satisfied exactly, so that

$$C = 0. \quad (\text{A.64})$$

Furthermore, if the constraint is satisfied initially, we would like it to remain satisfied during the evolution. In exact arithmetic this would imply

$$\partial_t C = 0. \quad (\text{A.65})$$

In a numerical evolution, however, truncation error and roundoff error generally produce small but nonzero values of C . It is therefore useful to reformulate the evolution system so that constraint violations propagate in a controlled way.

We now introduce the auxiliary variable

$$\Gamma \equiv D_i A^i. \quad (\text{A.66})$$

Taking a time derivative gives

$$\partial_t \Gamma = D_i \partial_t A^i. \quad (\text{A.67})$$

Using the evolution equation for A_i , we find

$$\partial_t \Gamma = -D_i E^i - D_i D^i \phi. \quad (\text{A.68})$$

Since $C = 0$ for constraint-satisfying data, we are free to add an arbitrary multiple of C to the right-hand side. Thus we write

$$\partial_t \Gamma = -D_i E^i - D_i D^i \phi + \eta C, \quad (\text{A.69})$$

where η is a constant. Using

$$C = D_i E^i - 4\pi\rho, \quad (\text{A.70})$$

this may also be written as

$$\partial_t \Gamma = (\eta - 1) D_i E^i - D_i D^i \phi - 4\pi\eta\rho. \quad (\text{A.71})$$

With this new variable, the Maxwell evolution system becomes

$$\partial_t A_i = -E_i - D_i \phi, \quad (\text{A.72})$$

$$\partial_t E_i = -D^2 A_i + D_i \Gamma - 4\pi j_i, \quad (\text{A.73})$$

$$\partial_t \Gamma = -D_i E^i - D_i D^i \phi + \eta C. \quad (\text{A.74})$$

Here $D^2 \equiv D_i D^i$. The advantage of this form is that the constraint violation no longer simply satisfies $\partial_t C = 0$. Instead, as we now show, it satisfies a wave equation.

Starting from

$$C = D_i E^i - 4\pi\rho, \quad (\text{A.75})$$

we take a time derivative:

$$\partial_t C = D_i \partial_t E^i - 4\pi \partial_t \rho. \quad (\text{A.76})$$

Using charge conservation,

$$\partial_t \rho = -D_i j^i, \quad (\text{A.77})$$

we obtain

$$\partial_t C = D_i \partial_t E^i + 4\pi D_i j^i. \quad (\text{A.78})$$

Substituting the evolution equation for E_i ,

$$\partial_t E_i = -D^2 A_i + D_i \Gamma - 4\pi j_i, \quad (\text{A.79})$$

gives

$$\partial_t C = D^i \left(-D^2 A_i + D_i \Gamma - 4\pi j_i \right) + 4\pi D_i j^i \quad (\text{A.80})$$

$$= -D^i D^2 A_i + D^i D_i \Gamma. \quad (\text{A.81})$$

Assuming that the spatial derivatives commute, this becomes

$$\partial_t C = -D^2 D^i A_i + D^2 \Gamma. \quad (\text{A.82})$$

Therefore,

$$\partial_t C = D^2 (\Gamma - D^i A_i). \quad (\text{A.83})$$

Taking another time derivative, we find

$$\partial_t^2 C = D^2 (\partial_t \Gamma - D^i \partial_t A_i). \quad (\text{A.84})$$

Using

$$\partial_t A_i = -E_i - D_i \phi, \quad (\text{A.85})$$

we have

$$-D^i \partial_t A_i = D^i E_i + D^i D_i \phi. \quad (\text{A.86})$$

Hence,

$$\partial_t \Gamma - D^i \partial_t A_i = (-D^i E_i - D^i D_i \phi + \eta C) + (D^i E_i + D^i D_i \phi) \quad (\text{A.87})$$

$$= \eta C. \quad (\text{A.88})$$

It follows that

$$\partial_t^2 C = D^2 (\eta C). \quad (\text{A.89})$$

Since η is constant, this reduces to

$$\partial_t^2 C = \eta D^2 C. \quad (\text{A.90})$$

Equivalently,

$$\boxed{(-\partial_t^2 + \eta D_i D^i) C = 0}. \quad (\text{A.91})$$

Thus the constraint violation C satisfies a wave equation. For $\eta > 0$, constraint violations propagate with speed $\sqrt{\eta}$. Analytically we haven't changed the nature of the equations but we have changed the numeric behavior as now the error will propagate a way as it turns out we will need to do something similar for the bssn equations.

BSSN Equations

We begin with the ADM equations:

$$R + K^2 - K_{ij} K^{ij} = 16\pi\rho, \quad \text{Hamiltonian constraint,} \quad (\text{A.92})$$

$$D_j (K^{ij} - \gamma^{ij} K) = 8\pi j^i, \quad \text{Momentum constraint,} \quad (\text{A.93})$$

$$\partial_t \gamma_{ij} = -2\alpha K_{ij} + \mathcal{L}_\beta \gamma_{ij}, \quad \text{Metric evolution,} \quad (\text{A.94})$$

$$\begin{aligned} \partial_t K_{ij} = & \alpha \left(R_{ij} - 2K_{ik} K^k_j + K K_{ij} \right) - D_i D_j \alpha \\ & - 4\pi\alpha \left[2S_{ij} - \gamma_{ij} (S - \rho) \right] + \mathcal{L}_\beta K_{ij}, \quad \text{Extrinsic-curvature evolution.} \end{aligned} \quad (\text{A.95})$$

The Ricci tensor can be written in several different forms. In particular, R_{ij} contains second derivatives of the metric:

$$R_{ij} = -\frac{1}{2}\gamma^{kl}(\partial_k\partial_l\gamma_{ij} + \partial_i\partial_j\gamma_{kl} - \partial_i\partial_l\gamma_{kj} - \partial_k\partial_j\gamma_{il}) + \gamma^{kl}(\Gamma^m_{il}\Gamma_{mkj} - \Gamma^m_{ij}\Gamma_{mkl}). \quad (\text{A.96})$$

The most important terms for rewriting the equations are the four terms involving second derivatives of the metric. The first term, $\gamma^{kl}\partial_k\partial_l\gamma_{ij}$, acts like a Laplace operator on the metric. However, because of the remaining second-derivative terms, we cannot immediately write the system in wave-equation form. In analogy with the introduction of an auxiliary variable in Maxwell's equations, we introduce a new variable here as well. This allows us to rewrite the Ricci tensor in terms of the trace of the Christoffel symbols. In particular, we define

$$\Gamma^i \equiv \gamma^{jk}\Gamma^i_{jk}. \quad (\text{A.97})$$

In terms of this contracted Christoffel symbol, the Ricci tensor can be written as

$$\begin{aligned} R_{ij} = & -\frac{1}{2}\gamma^{kl}\partial_k\partial_l\gamma_{ij} + \gamma_{k(i}\partial_{j)}\Gamma^k + \Gamma^k\Gamma_{(ij)k} \\ & + 2\gamma^{lm}\Gamma^k_{m(i}\Gamma_{j)kl} + \gamma^{kl}\Gamma^m_{il}\Gamma_{mkj}. \end{aligned} \quad (\text{A.98})$$

We can also use the identity relating the contracted Christoffel symbols to the divergence of the inverse metric:

$$g^{bc}\Gamma^a_{bc} = -\frac{1}{\sqrt{|g|}}\partial_b(\sqrt{|g|}g^{ab}). \quad (\text{A.99})$$

Applying this identity to the spatial metric gives

$$\Gamma^i = -\frac{1}{\sqrt{\gamma}}\partial_j(\sqrt{\gamma}\gamma^{ij}). \quad (\text{A.100})$$

We can now regard Γ^i as an independent variable. It is related to the divergence of the metric in much the same way as the auxiliary variable in the Maxwell example above. To complete the system, we must derive an evolution equation for Γ^i . This is done by taking a time derivative of Eq. (A.100). Doing so introduces divergences of the extrinsic curvature. These terms can then be simplified using the momentum constraint.

To avoid some of the difficulties associated with the determinant of the metric, we also introduce a conformal transformation. In particular, we write

$$\gamma_{ij} = e^{4\phi} \bar{\gamma}_{ij}. \quad (\text{A.101})$$

There is some freedom in the choice of conformal factor. We can then rewrite quantities such as the Christoffel symbols and the Ricci tensor in terms of the conformal metric plus correction terms involving the conformal factor. The physical Ricci tensor can be split into two pieces,

$$R_{ij} = \bar{R}_{ij} + R_{ij}^\phi, \quad (\text{A.102})$$

where $\bar{R}_{ij}$ is computed from the conformally related metric $\bar{\gamma}_{ij}$, and R_{ij}^ϕ contains the terms involving the conformal factor ϕ .

The conformal Ricci tensor is

$$\begin{aligned} \bar{R}_{ij} = & -\frac{1}{2} \bar{\gamma}^{kl} \partial_k \partial_l \bar{\gamma}_{ij} + \bar{\gamma}_{k(i} \partial_{j)} \bar{\Gamma}^k + \bar{\Gamma}^k \bar{\Gamma}_{(ij)k} \\ & + 2 \bar{\gamma}^{lm} \bar{\Gamma}_{m(i} \bar{\Gamma}_{j)kl} + \bar{\gamma}^{kl} \bar{\Gamma}_{il}^m \bar{\Gamma}_{mkj}. \end{aligned} \quad (\text{A.103})$$

The conformal-factor contribution is

$$\begin{aligned} R_{ij}^\phi = & -2 \left(\bar{D}_i \bar{D}_j \phi + \bar{\gamma}_{ij} \bar{\gamma}^{lm} \bar{D}_l \bar{D}_m \phi \right) \\ & + 4 \left((\bar{D}_i \phi) (\bar{D}_j \phi) - \bar{\gamma}_{ij} \bar{\gamma}^{lm} (\bar{D}_l \phi) (\bar{D}_m \phi) \right). \end{aligned} \quad (\text{A.104})$$

The contracted conformal Christoffel symbols are defined by

$$\bar{\Gamma}^i \equiv \bar{\gamma}^{jk} \bar{\Gamma}_{jk}^i = -\frac{1}{\bar{\gamma}^{1/2}} \partial_j \left(\bar{\gamma}^{1/2} \bar{\gamma}^{ij} \right), \quad (\text{A.105})$$

where $\bar{\gamma}$ denotes the determinant of the conformal metric $\bar{\gamma}_{ij}$.

Using the freedom in the choice of conformal factor, we may impose

$$\bar{\gamma} = 1, \quad (\text{A.106})$$

which reduces the conformal connection functions to

$$\bar{\Gamma}^i = -\partial_j \bar{\gamma}^{ij}. \quad (\text{A.107})$$

Taking the determinant of

$$\gamma_{ij} = e^{4\phi} \bar{\gamma}_{ij}, \quad (\text{A.108})$$

and using $\bar{\gamma} = 1$, we obtain

$$\phi = \frac{1}{12} \ln \gamma, \quad (\text{A.109})$$

where γ is the determinant of the physical metric γ_{ij} .

The extrinsic curvature is decomposed into its trace and trace-free parts:

$$K_{ij} = A_{ij} + \frac{1}{3} \gamma_{ij} K, \quad (\text{A.110})$$

where

$$K = \gamma^{ij} K_{ij}. \quad (\text{A.111})$$

Finally, the trace-free part is conformally rescaled according to

$$A_{ij} = e^{4\phi} \tilde{A}_{ij}. \quad (\text{A.112})$$

The variables we evolve are therefore

$$\phi, \quad \tilde{\gamma}_{ij}, \quad \tilde{A}_{ij}, \quad K, \quad \tilde{\Gamma}^i.$$

These are the standard BSSN variables.

The BSSN equations are

$\partial_t \phi = -\frac{1}{6} \alpha K + \beta^i \partial_i \phi + \frac{1}{6} \partial_i \beta^i,$	Conformal exponent evolution,
$\partial_t K = -\gamma^{ij} D_i D_j \alpha + \alpha \left(\tilde{A}_{ij} \tilde{A}^{ij} + \frac{1}{3} K^2 \right) + 4\pi \alpha (\rho + S) + \beta^i \partial_i K,$	Mean-curvature evolution,
$\partial_t \tilde{\gamma}_{ij} = -2\alpha \tilde{A}_{ij} + \beta^k \partial_k \tilde{\gamma}_{ij} + \tilde{\gamma}_{ik} \partial_j \beta^k + \tilde{\gamma}_{kj} \partial_i \beta^k - \frac{2}{3} \tilde{\gamma}_{ij} \partial_k \beta^k,$	Conformal-metric evolution,
$\partial_t \tilde{A}_{ij} = e^{-4\phi} \left[- (D_i D_j \alpha)^{\text{TF}} + \alpha \left(R_{ij}^{\text{TF}} - 8\pi S_{ij}^{\text{TF}} \right) \right] + \alpha \left(K \tilde{A}_{ij} - 2 \tilde{A}_{il} \tilde{A}^l{}_j \right) + \beta^k \partial_k \tilde{A}_{ij} + \tilde{A}_{ik} \partial_j \beta^k + \tilde{A}_{kj} \partial_i \beta^k - \frac{2}{3} \tilde{A}_{ij} \partial_k \beta^k,$	Trace-free curvature evolution,
$\partial_t \tilde{\Gamma}^i = -2 \tilde{A}^{ij} \partial_j \alpha + 2\alpha \left(\tilde{\Gamma}^i{}_{jk} \tilde{A}^{kj} - \frac{2}{3} \tilde{\gamma}^{ij} \partial_j K - 8\pi \tilde{\gamma}^{ij} j_j + 6 \tilde{A}^{ij} \partial_j \phi \right) + \beta^j \partial_j \tilde{\Gamma}^i - \tilde{\Gamma}^j \partial_j \beta^i + \frac{2}{3} \tilde{\Gamma}^i \partial_j \beta^j + \frac{1}{3} \tilde{\gamma}^{li} \partial_l \partial_j \beta^j + \tilde{\gamma}^{lj} \partial_j \partial_l \beta^i,$	Conformal-connection evolution.

(A.113)

Gauge and Slicing Conditions

We are not quite done, because we still have freedom to choose our coordinate system. In particular, we must choose conditions for the lapse α and the shift β^i . We summarize several common choices below.

Geodesic slicing. A simple choice is to set

$$\alpha = 1, \quad \beta^i = 0. \quad (\text{A.114})$$

This is called geodesic slicing. Although this is an easy choice, it is usually a poor one for numerical relativity simulations. The acceleration of the normal observers is

$$a_a = D_a \ln \alpha. \quad (\text{A.115})$$

For geodesic slicing, $\alpha = 1$, so $a_a = 0$. Thus the normal observers are in free fall. This is also true for the coordinate points. In a strong gravitational field, this can lead to coordinate singularities, making geodesic slicing unsuitable for most numerical simulations of black holes.

Maximal slicing. We can control the convergence of coordinate observers by controlling the mean curvature. In maximal slicing, we impose

$$K = 0. \quad (\text{A.116})$$

We have imposed this condition on initial data in the past, but here we impose it at all times. Therefore,

$$\partial_t K = 0. \quad (\text{A.117})$$

This condition has several desirable properties, including singularity avoidance. In particular, the lapse tends to collapse toward zero near strong-field regions. This causes those slices to be held back in the evolution, a phenomenon known as the collapse of the lapse. However, maximal slicing also has disadvantages, such as grid stretching, which can lead to sharp features that are difficult to resolve numerically.

Harmonic coordinates. Harmonic coordinates are defined by the condition

$$\Gamma^a = 0. \quad (\text{A.118})$$

A related condition is harmonic slicing, in which we require only the time component of Γ^a to vanish and set $\beta^i = 0$. To obtain a lapse condition, we compute

$$\Gamma^0 = -\frac{1}{\alpha\gamma^{1/2}}\partial_t(\gamma^{1/2}\alpha^{-1}) = 0. \quad (\text{A.119})$$

This gives the lapse evolution equation

$$\partial_t \alpha = -\alpha^2 K. \quad (\text{A.120})$$

Solutions to the resulting wave equation are generally well behaved. However, with a few modifications to harmonic slicing, we can obtain an even better gauge choice for black-hole evolutions.

1 + log slicing and the Gamma-driver condition. We generalize the lapse evolution equation to

$$\partial_t \alpha = -\alpha^2 f(\alpha) K. \quad (\text{A.121})$$

For $f(\alpha) = 1$, this reduces to harmonic slicing. For $f(\alpha) = 0$, the lapse is constant in time, which gives geodesic slicing when $\alpha = 1$. In the limit $f(\alpha) \rightarrow \infty$, this condition approaches maximal slicing.

A particularly useful choice is

$$f(\alpha) = \frac{2}{\alpha}. \quad (\text{A.122})$$

This gives

$$\partial_t \alpha = -2\alpha K, \quad (\text{A.123})$$

which is the standard 1 + log slicing condition. This choice has the singularity-avoidance properties of maximal slicing while retaining good behavior in nearly flat regions.

As a further generalization, we do not assume that β^i is close to zero. We include the advective term and write

$$(\partial_t - \beta^i \partial_i) \alpha = -\alpha^2 f(\alpha) K. \quad (\text{A.124})$$

For $f(\alpha) = 2/\alpha$, this becomes

$$(\partial_t - \beta^i \partial_i) \alpha = -2\alpha K. \quad (\text{A.125})$$

We also need an evolution condition for the shift. A common choice is the Gamma-driver condition,

$$(\partial_t + \beta^j \partial_j) \beta^i = \mu B^i, \quad (\text{A.126})$$

together with

$$(\partial_t + \beta^j \partial_j) B^i = (\partial_t + \beta^j \partial_j) \tilde{\Gamma}^i - \eta B^i. \quad (\text{A.127})$$

Here μ and η are constants. The combination of 1 + log slicing and the Gamma-driver shift condition is called the moving-puncture gauge. A major advantage of this gauge is that it removes the need for excision: the singularities are effectively handled by the gauge itself.

Appendix B

EMDA Black Holes

Appendix C

Computation of Ψ_4

C.1 Computation of Ψ_4

The gravitational waveforms extracted from numerical relativity simulations are often expressed in terms of the Newman–Penrose scalar Ψ_4 . This scalar provides a convenient way to isolate the outgoing gravitational radiation from the curvature of spacetime. In this appendix, we describe the construction of Ψ_4 , its relation to the Weyl tensor, and its connection to the gravitational-wave strain measured by detectors.

C.1.1 Tetrad Construction

We begin by constructing an orthonormal tetrad adapted to the spatial hypersurfaces. We take the timelike vector to be the future-directed unit normal n^a to the spacelike hypersurface. In the asymptotic region, we construct a set of spatial vectors resembling the spherical-coordinate directions and then orthonormalize them using the Gram–Schmidt procedure.

We start with the spatial vectors

$$\begin{aligned} u^a &= (0, x, y, z), \\ v^a &= (0, xz, yz, -x^2 - y^2), \\ w^a &= (0, -y, x, 0). \end{aligned} \tag{C.1}$$

Using the spatial metric γ_{ij} , we define the spatial inner product and norm by

$$\langle u, v \rangle \equiv \gamma_{ij} u^i v^j, \tag{C.2}$$

and

$$\|u\| \equiv \sqrt{\langle u, u \rangle}. \tag{C.3}$$

The Gram–Schmidt procedure then gives the orthonormal spatial vectors

$$\begin{aligned} e_{(1)}^a &= \frac{u^a}{\|u\|}, \\ e_{(2)}^a &= \frac{v^a - \langle e_{(1)}, v \rangle e_{(1)}^a}{\|v - \langle e_{(1)}, v \rangle e_{(1)}\|}, \\ e_{(3)}^a &= \frac{w^a - \langle e_{(1)}, w \rangle e_{(1)}^a - \langle e_{(2)}, w \rangle e_{(2)}^a}{\|w - \langle e_{(1)}, w \rangle e_{(1)} - \langle e_{(2)}, w \rangle e_{(2)}\|}. \end{aligned} \quad (\text{C.4})$$

We identify these vectors with the radial, polar, and azimuthal directions,

$$e_{(1)}^a = r^a, \quad e_{(2)}^a = \theta^a, \quad e_{(3)}^a = \phi^a. \quad (\text{C.5})$$

Using these orthonormal vectors, we construct the null tetrad

$$\begin{aligned} \ell^a &= \frac{1}{\sqrt{2}} (n^a + r^a), \\ k^a &= \frac{1}{\sqrt{2}} (n^a - r^a), \\ m^a &= \frac{1}{\sqrt{2}} (\theta^a + i\phi^a), \\ \bar{m}^a &= \frac{1}{\sqrt{2}} (\theta^a - i\phi^a). \end{aligned} \quad (\text{C.6})$$

Here ℓ^a and k^a are real null vectors, while m^a is a complex null vector and $\bar{m}^a$ is its complex conjugate. With the metric signature $(-, +, +, +)$, the tetrad satisfies

$$\ell^a \ell_a = 0, \quad k^a k_a = 0, \quad m^a m_a = 0, \quad \bar{m}^a \bar{m}_a = 0, \quad (\text{C.7})$$

and the only non-vanishing inner products are

$$\ell^a k_a = -1, \quad m^a \bar{m}_a = 1. \quad (\text{C.8})$$

This null tetrad forms the basis used in the Newman–Penrose formalism.

C.1.2 The Weyl Tensor and Ψ_4

The Weyl tensor is the trace-free part of the spacetime curvature. In four dimensions, the Riemann tensor has twenty independent components. Ten of these are determined by the Ricci tensor, while the remaining ten are contained in the Weyl tensor. The Weyl tensor therefore captures the part of the curvature associated with tidal and radiative degrees of freedom.

In n dimensions, with $n \geq 4$, the Weyl tensor is defined by

$$C_{abcd} = R_{abcd} + \frac{1}{n-2} (g_{ad}R_{cb} + g_{bc}R_{da} - g_{ac}R_{db} - g_{bd}R_{ca}) \\ + \frac{1}{(n-1)(n-2)} (g_{ac}g_{db} - g_{ad}g_{cb}) R. \quad (\text{C.9})$$

In four dimensions, this becomes

$$C_{abcd} = R_{abcd} + \frac{1}{2} (g_{ad}R_{cb} + g_{bc}R_{da} - g_{ac}R_{db} - g_{bd}R_{ca}) \\ + \frac{1}{6} (g_{ac}g_{db} - g_{ad}g_{cb}) R. \quad (\text{C.10})$$

The Weyl tensor has the same index symmetries as the Riemann tensor:

$$C_{abcd} = -C_{abdc} = -C_{bacd} = C_{cdab}, \quad (\text{C.11})$$

together with the cyclic identity

$$C_{abcd} + C_{adbc} + C_{acdb} = 0. \quad (\text{C.12})$$

The Newman–Penrose scalar Ψ_4 is defined as a projection of the Weyl tensor onto the null tetrad:

$$\Psi_4 = -C_{abcd} k^a \bar{m}^b k^c \bar{m}^d. \quad (\text{C.13})$$

The overall sign convention may vary between authors and numerical relativity codes. In an asymptotically flat spacetime, with an appropriate outgoing tetrad, Ψ_4 encodes the outgoing gravitational radiation.

C.1.3 Relation to the 3 + 1 Variables

In numerical relativity, Ψ_4 is computed from quantities defined on spatial hypersurfaces. The curvature terms are therefore rewritten using the Gauss–Codazzi relations. Schematically, the relevant projections are

$$\perp R_{abcd} = {}^{(3)}R_{abcd} + K_{ac}K_{bd} - K_{ad}K_{bc}, \quad (\text{C.14})$$

$$\perp (R_{abcd}n^a) = D_c K_{bd} - D_d K_{bc}, \quad (\text{C.15})$$

and

$$\perp (R_{abcd}n^a n^c) = \mathcal{L}_n K_{bd} + K_{bc}K^c_d + \frac{1}{\alpha} D_b D_d \alpha. \quad (\text{C.16})$$

These identities allow the spacetime curvature entering Eq. (C.13) to be expressed in terms of the spatial metric γ_{ij} , the extrinsic curvature K_{ij} , and their spatial derivatives.

After substituting the Gauss–Codazzi relations, the Weyl scalar can be written in terms of spatial curvature and extrinsic-curvature quantities. A common form is

$$\begin{aligned} \Psi_4 = & \left({}^{(3)}R_{bd} + K K_{bd} - K_{bc}K^c_d \right) \bar{m}^b \bar{m}^d \\ & + (D_c K_{bd} - D_d K_{bc}) r^c \bar{m}^b \bar{m}^d. \end{aligned} \quad (\text{C.17})$$

This expression shows explicitly that Ψ_4 can be computed from the spatial geometry and extrinsic curvature of each slice.

For BSSN-type formulations, the extrinsic curvature is decomposed as

$$K_{ij} = \chi^{-1} \left(\tilde{A}_{ij} + \frac{1}{3} \tilde{\gamma}_{ij} K \right), \quad (\text{C.18})$$

where $\tilde{\gamma}_{ij}$ is the conformal spatial metric, $\tilde{A}_{ij}$ is the trace-free conformal extrinsic curvature, K is the trace of the extrinsic curvature, and χ is the conformal factor used in the evolution system.

The resulting expressions for the real and imaginary parts of Ψ_4 in terms of BSSN variables are lengthy. They are typically evaluated directly in the code using the evolved variables and their

derivatives. Symbolically, one may write

$$\Psi_4 = \mathcal{R}(\Psi_4) + i\mathcal{I}(\Psi_4), \quad (\text{C.19})$$

where the real and imaginary parts are obtained by projecting the spatial curvature and extrinsic-curvature terms onto the complex dyad formed from m^a and $\bar{m}^a$.

C.1.4 Relation Between Ψ_4 and Strain

The quantity directly measured by gravitational-wave detectors such as LIGO is not Ψ_4 , but the gravitational-wave strain. The strain is commonly written as the complex quantity

$$h = h_+ - ih_\times, \quad (\text{C.20})$$

where h_+ and $h_\times$ are the two gravitational-wave polarizations.

At future null infinity, Ψ_4 is related to the second time derivative of the strain by

$$\Psi_4 = \ddot{h}_+ - i\ddot{h}_\times = \ddot{h}. \quad (\text{C.21})$$

Therefore, the strain can be obtained by integrating Ψ_4 twice in time:

$$h(t) = \int^t dt' \int^{t'} dt'' \Psi_4(t''). \quad (\text{C.22})$$

In practice, this integration must be performed carefully, since direct time-domain integration can introduce secular drifts. Frequency-domain methods or fixed-frequency integration are often used to control these low-frequency artifacts.

The strain and Weyl scalar may also be decomposed into spin-weighted spherical harmonics:

$$h(t, \theta, \phi) = \sum_{\ell, m} h_{\ell m}(t) {}_{-2}Y_{\ell m}(\theta, \phi), \quad (\text{C.23})$$

$$\Psi_4(t, \theta, \phi) = \sum_{\ell, m} \Psi_4^{\ell m}(t) {}_{-2}Y_{\ell m}(\theta, \phi). \quad (\text{C.24})$$

For each multipolar mode, the relation between Ψ_4 and the strain becomes

$$\Psi_4^{\ell m}(t) = \ddot{h}_{\ell m}(t), \quad (\text{C.25})$$

or equivalently,

$$h_{\ell m}(t) = \int^t dt' \int^{t'} dt'' \Psi_4^{\ell m}(t''). \quad (\text{C.26})$$

Often, waveform comparisons focus on the dominant $(\ell, m) = (2, \pm 2)$ modes.

C.1.5 Waveform Mismatch

To quantify how similar two gravitational waveforms are, we compute a mismatch. This can be done in two related ways. A purely numerical comparison measures the difference between two waveforms directly, while a detector-based comparison weights the difference by the noise curve of a gravitational-wave detector.

For two complex waveforms $h_1(t)$ and $h_2(t)$, a simple normalized overlap may be written as

$$\mathcal{O} = \frac{|\langle h_1, h_2 \rangle|}{\sqrt{\langle h_1, h_1 \rangle \langle h_2, h_2 \rangle}}, \quad (\text{C.27})$$

where the time-domain inner product may be taken as

$$\langle h_1, h_2 \rangle = \int_{t_1}^{t_2} h_1(t) h_2^*(t) dt. \quad (\text{C.28})$$

The mismatch is then

$$\mathcal{M} = 1 - \mathcal{O}. \quad (\text{C.29})$$

In practice, the overlap is usually maximized over arbitrary time and phase shifts:

$$\mathcal{M} = 1 - \max_{\Delta t, \Delta \phi} \frac{\left| \langle h_1, h_2^{\Delta t, \Delta \phi} \rangle \right|}{\sqrt{\langle h_1, h_1 \rangle \langle h_2, h_2 \rangle}}. \quad (\text{C.30})$$

This provides a single number measuring the disagreement between two waveforms after removing trivial offsets in arrival time and phase. A smaller mismatch indicates better agreement.

Appendix D

Dendro Notes

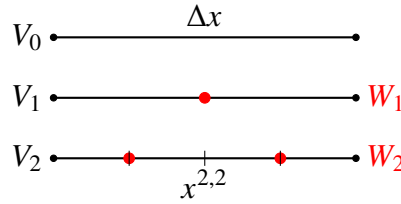
D.1 Wavelet adaptive mesh refinement

We can construct a sparse representation of the fields using a set of nested dyadic grids:

$$V_j = \{x^{j,k} = 2^{-j} k \Delta x\}, \quad (\text{D.1})$$

where V_j denotes the grid level, and $x^{j,k}$ denotes the points on the grids.

I show an example below: the red dots indicate the points included only in the alternative grids W_j . Notice that all points in a given V_j will also appear in all subsequent refinement levels. The grids do not need to separately save the points already included at coarser levels; the points unique to finer grids are highlighted in red.



We start with a field defined on the points of the grid. The field is represented by $u^{j,k}$. We can extend the values to higher levels by interpolation. For points already included at coarser resolutions, we have $u^{j+1,2k} = u^{j,k}$. For new points, interpolation gives

$$u^{j+1,2k+1} = \sum_m h_{j,m}^{j+1,2k+1} u^{j,m}. \quad (\text{D.2})$$

We can represent this interpolation with a series of basis functions:

$$\phi_{j,k}(x) = \sum_l c^l \phi_{j+1,l}(x), \quad (\text{D.3})$$

where c^l are the interpolation weights.

To represent the full grid, including the alternative grids W_j (the red dots above), we can write:

$$u(x) = \sum_{k \in S_0} u^{0,k} \phi_{0,k}(x) + \sum_{j=1} \sum_{k \in S_j} d^{j,k} \phi_{j,k}(x), \quad (\text{D.4})$$

where $d^{j,k}$ are the wavelet coefficients. The $u^{0,k}$ values are the field values at base level points, $u^{0,k} = u(x^{0,k})$, and $u^{j,k} = u(x^{j,k})$.

The wavelet coefficients are computed by comparing the interpolation from the previous level to the actual field value:

$$d^{j,k} = u^{j,k} - P(x^{j,k}, j-1), \quad (\text{D.5})$$

where $P(x^{j,k}, j-1)$ is the interpolated value from level $j-1$.

The wavelet coefficient measures how well the interpolation captures the field at that point. We can set a threshold ϵ such that refinement occurs only where

$$|d^{j,k}| > \epsilon.$$

D.2 Spacefilling curves and the octree

Appendix E

CFD Notes

Appendix F

EMDA Code Setup on MaryLou

F.1 Building `emda-gr`

To build `emda-gr`, the following external packages are required:

- **C/C++11 or higher** (tested with GNU and Intel compilers)
- **MPI implementation** (tested with MVAPICH2, OpenMPI, and Intel MPI)
- **BLAS/LAPACK** (tested with OpenBLAS and Intel MKL)
- **GSL** (GNU Scientific Library); set `GSL_ROOT_DIR` in CMake to enable automatic detection

The modules currently loaded on MaryLou are:

```
andrewc0@login03:$ ml
```

Currently Loaded Modules:

- 1) gcc/14.1
- 2) python/3.12
- 3) cmake/3.28

- 4) gcc-runtime/13.2.0-g4hc7ev
- 5) eigen/3.4.0-6kqvial
- 6) openmpi5-GR
- 7) gsl-GR
- 8) mkl-GR
- 9) rclone/1.65.2-tyg4lvy

To ensure these dependencies are always available, it is convenient to load them automatically in your shell configuration. Open your `~/.bash_profile`:

```
vim ~/.bash_profile
```

An example `.bash_profile` is shown below. Additional customizations may be added as needed.

```
# --- Modules ---
module purge
module use /grphome/grp_GR/.modulefiles
module load gcc/14.1 python/3.12 cmake/3.28 eigen/3.4.0-6kqvial
module load openmpi5-GR gsl-GR mkl-GR rclone

# --- GSL ---
export GSL_DIR=/grphome/grp_GR/apps/gsl-2.8
export GSL_ROOT_DIR=/grphome/grp_GR/apps/gsl-2.8
export LD_LIBRARY_PATH="${GSL_DIR}/lib:${LD_LIBRARY_PATH:-}"

# --- MKL ---
if [ -n "${MKLROOT:-}" ] && [ -d "${MKLROOT}/lib/intel64" ]; then
```

```
export LD_LIBRARY_PATH="${MKLRROOT}/lib/intel64:${LD_LIBRARY_PATH:-}"
fi

# --- GCC 14 runtime (libstdc++, libgcc_s) ---
# Required to avoid GLIBCXX / CXXABI runtime errors
if command -v g++ >/dev/null 2>&1; then
    GCC_LIBDIR="$(dirname "$(g++ -print-file-name=libstdc++.so.6)")"
    GCC_GCCLIBDIR="$(dirname "$(g++ -print-file-name=libgcc_s.so.1)")"

    if [ -d "$GCC_LIBDIR" ]; then
        export LD_LIBRARY_PATH="$GCC_LIBDIR:${LD_LIBRARY_PATH:-}"
    fi

    if [ -d "$GCC_GCCLIBDIR" ]; then
        export LD_LIBRARY_PATH="$GCC_GCCLIBDIR:${LD_LIBRARY_PATH:-}"
    fi
fi

# --- History settings ---
HISTSIZE=1000000
HISTFILESIZE=2000000
HISTCONTROL=ignoredups:erasedups
HISTIGNORE="ls:bg:fg:history"
shopt -s histappend

# Save history after every command
```

```
PROMPT_COMMAND="history -a; history -c; history -r; $PROMPT_COMMAND"
```

After reloading your shell, verify that the modules have been loaded correctly:

```
m1
```

The output should match the module list shown above.

F.2 Building the Solver

First, clone the EMDA repository. Create a project directory if desired, then run:

```
git clone git@github.com:dfvankomen/emda-gr.git
cd emda-gr
```

CMake will now automatically fetch `dendrolib`.

Create a build directory:

```
mkdir build
cd build
```

You may then configure the project using:

```
ccmake ..
```

CMake will locate the required libraries and present a configuration interface. After configuration, generate the build files. At present, the only settings you may wish to toggle is the choice between the EMDA-EMD-EMGR systems of equations. The second is the CPU architecture we will be running on. For the purposes of the first run, let us use EMDA.

We will also need to specify the CPU architecture for the target system. For example on MaryLou, we normally use the `physics2` or `m12` partitions, both of which consist of 64-core AMD EPYC

7763 (2.45 GHz) CPUs. Therefore, we should build for the Zen 3 architecture by setting `CPU_ARCH` to `znver3`.

Once the build files are generated, compile the code using:

```
make -j
```

F.2.1 Build Outputs

Once the build completes, you should have both `tpid` and `emdaSolver` binaries located in one of the following directories:

```
emda-gr/build/emdaSolver/
```

At this point, the solver is ready, and we can proceed to running simulations.

F.3 Runs and Job Scripts

With the solver built, we now need to move from your home directory into the scratch directory. The scratch directory varies by supercomputer. Consult the Supercomputer documentation I will show how this is done on Marylou.

First, change into your scratch area:

```
cd /nobackup/autodelete/usr/YOURUSERNAME/
```

Create a directory for your runs (name it however you like), then enter it:

```
mkdir EMDA_runs
```

```
cd EMDA_runs
```

Next, copy over the binaries we just built, along with a parameter (TOML) file. (The ~ in these commands refers to your home directory.)

Copy emdaSolver:

```
cp ~/path/to/emda-gr/build/build_noavx/emdaSolver/emdaSolver ./
```

Copy tpid:

```
cp ~/path/to/emda-gr/build/build_noavx/emdaSolver/tpid ./
```

Copy the parameter file:

```
cp ~/path/to/emda-gr/emda_simplified.param.toml ./
```

Verify that everything copied correctly:

```
ls
```

You should see:

```
emda_simplified.param.toml  emdaSolver  tpid
```

Now create output directories for the run:

```
mkdir cp prof vtu bah
```

The vtu is where visual data that we can see in paraview is saved, cp is where checkpoints are saved prof is where the gravitational and other waves are saved as well as the ADM Mass charge and the constraint solutions are saved. (We will add another section later describing the relevant parameter-file settings.)

F.3.1 Submitting a Run with Slurm

To run on the cluster, create a job script. Use vim:

```
vim Job.sb
```

Paste the following:

```
#!/bin/zsh

#SBATCH --time=72:00:00

#SBATCH --ntasks=256

#SBATCH --nodes=2

#SBATCH --mem-per-cpu=4096M

#SBATCH -J EMDA_RUN

#SBATCH --mail-user=youremail@email.com

#SBATCH --mail-type=END,FAIL

#SBATCH --partition=m12


echo "Starting run"


source ~/.bash_profile

./tpid emda_simplified.param.toml 256 &> TPID.txt

mpirun -np 256 ./emdaSolver emda_simplified.param.toml &>> Inspiral.txt


echo "Run completed"
```

Note that the semicolon means “do this before moving on.” We need this behavior so that the TPID solver finishes generating the initial data before proceeding. TPID will produce a binary file called `rit_q2_tpid_sol.bin`, which `emdaSolver` will then read and evolve.

On the m12 nodes, each node has 128 CPU cores, so you should scale the number of nodes accordingly (e.g., 256 tasks typically corresponds to 2 nodes).

You can submit the job using:

```
sbatch Job.sb
```

Once submitted, the job will enter the queue and will begin running when resources become available. You can check the status of your jobs with:

```
squeue | grep yourusername
```

After the job starts running, you can monitor the output by inspecting the text files generated during execution, for example:

```
cat TPID.txt
```

And that's it—the simulation is running.

The only remaining step is to explain the parameter file:

F.4 Par-file Explanation

The parameter file is read by both the `tpid` solver and `emdaSolver`. It allows us to change solver settings without having to rebuild the code each time, which is extremely useful.

Below is an overview of the most important settings. If a parameter is not mentioned here, it is probably safe to leave it at its default value.

```
# Whether or not the solver should attempt to restore from a checkpoint  
EMDA_RESTORE_SOLVER = false
```

This determines whether the solver will attempt to pick up where it left off by restoring from a checkpoint.

```
EMDA_IO_OUTPUT_FREQ = 400
```

This controls how often VTU output is written. Smaller values produce output more frequently but increase storage usage. Adjust this setting to suit your needs.

```
EMDA_REMESH_TEST_FREQ = 200
```

This sets how often the solver invokes the AMR algorithm to determine whether the grid resolution needs to change. Setting this too often can add noise on the grid. updating too infrequently can mean the dynamics are not well captured by the grid. Around 200 seem to work well.

```
EMDA_GW_EXTRACT_FREQ = 25
```

This controls how often gravitational waves are extracted, along with black hole positions and masses. This step is not particularly expensive, either computationally or in terms of storage, so values in the range 50–100 are typically reasonable.

```
EMDA_CHECKPT_FREQ = 100
```

This specifies the number of time steps between checkpoints. This is pretty cheap computationally so this can be small.

```
# VTU file prefix
```

```
EMDA_VTU_FILE_PREFIX = "vtu/emda_gr"
```

```
# Checkpoint file prefix
```

```
EMDA_CHKPT_FILE_PREFIX = "cp/emda_cp"
```

```
# Profiling file prefix
```

```
EMDA_PROFILE_FILE_PREFIX = "prof/emda_prof"
```

These options control where different types of output data are stored.

```
# Which evolution variables to output
EMDA_VTU_OUTPUT_EVOL_INDICES = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14,
15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34,
35]
```

```
# Number of evolution variables to output
EMDA_NUM_EVOL_VARS_VTU_OUTPUT = 36
```

These settings determine which evolution variables (e.g., metric components, extrinsic curvature, etc.) are written to the VTU files. If you need to reduce storage usage, you may omit some of these variables.

```
# Which constraint variables to output
EMDA_VTU_OUTPUT_CONST_INDICES = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13]

# Number of constraint variables to output
EMDA_NUM_CONST_VARS_VTU_OUTPUT = 14
```

These are similar to the evolution-variable settings, but apply to constraint quantities such as the Hamiltonian and momentum constraints.

```
# Whether to isolate an X slice
EMDA_VTU_X_SLICE = false

# Whether to isolate a Y slice
EMDA_VTU_Y_SLICE = false

# Whether to isolate a Z slice
EMDA_VTU_Z_SLICE = true

# Any combination may be set to true
```

In general, writing VTU output for the entire 3-D grid is expensive, so we typically output a 2-D slice instead. You should enable only one of these slice options at a time; enabling multiple slices will produce unintuitive cuts through the grid. For most purposes, leaving the Z slice enabled is a good choice.

```
EMDA_DENDRO_GRAIN_SZ = 50
```

This controls how many grid blocks are assigned to each processor. Some analysis has been done to estimate optimal values, but in practice this parameter should generally be kept between 25 and 200.

```
EMDA_DENDRO_AMR_FAC = 0.2
```

This controls how sensitive the AMR algorithm is to refining or coarsening the grid.

```
EMDA_ELE_ORDER = 6
```

Loosely speaking, this determines how much information each block shares with its neighbors. Higher values increase communication and can slow the code down. For EMDA, this should currently be left at 6.

```
# Minimum depth of the mesh/octree
```

```
EMDA_MINDEPTH = 4
```

```
# Maximum depth of the mesh/octree
```

```
EMDA_MAXDEPTH = 15
```

These are very important parameters. They determine how coarse (EMDA_MINDEPTH) and how fine (EMDA_MAXDEPTH) the grid is allowed to become. Note that the CFL factor is set by the finest level of the grid, so increasing EMDA_MAXDEPTH significantly increases the runtime. Each increase

in EMDA_MAXDEPTH halves the grid spacing at the finest level. Each black hole needs a minimum of 50 or more points accross the horizon to be adequately resolved. Maxdepth 15 for a spaceteim with a total mass of 1, with an equal mass ratio merger on a ± 400 grid is adequately resolved. as q goes up this increases for a q=2 you need a maxdepth of 16.

```
#####
```

```
# REFINEMENT STRATEGY
```

```
# Which refinement mode to use
```

```
# 0 - WAMR
```

```
# 1 - EH
```

```
# 2 - EH_WAMR
```

```
# 3 - BH_LOC
```

```
# 4 - BH_WAMR
```

```
EMDA_REFINEMENT_MODE = 4
```

This parameter controls how the grid is refined. it is recommended to leave this set to 4 for now.

```
EMDA_AMR_R_RATIO = 1.618033988749895
```

This determines how quickly refinement around a black hole falls off with radius. The default value is reasonable and should be left unchanged initially.

```
# Wavelet Tolerance Function to use (see grUtils.cpp for adding more)
```

```
# 1,2 - deprecated
```

```
# 3 - center highly refined; low refinement elsewhere eases over time
```

```
# 4 - start with refinement centered at BHs, then widen to r=0 focused
```

```
# 5 - focus refinement in self-similar ways about each BH
```

```
# 6 - refine well in  $r \leq t$  and coarsely in  $r > t$   
EMDA_USE_WAVELET_TOL_FUNCTION = 6
```

This controls how the refinement tolerance varies across the grid. Function 6, which was developed by William, works well in practice and is recommended. More details on this can be found [52]

```
# Minimum wavelet tolerance  
EMDA_WAVELET_TOL = 0.00001
```

```
# Gravitational wave refinement tolerance  
EMDA_GW_REFINE_WTOL = 0.0001
```

```
# Maximum wavelet tolerance  
EMDA_WAVELET_TOL_MAX = 0.001
```

```
# Wavelet tolerance radii (for shell-based tolerance)  
EMDA_WAVELET_TOL_FUNCTION_R0 = 3.0  
EMDA_WAVELET_TOL_FUNCTION_R1 = 12.0
```

The minimum wavelet tolerance is the most important of these parameters, as it sets the strictest refinement criterion over the entire grid. The remaining parameters depend on the chosen tolerance function; the defaults are reasonable, but may be adjusted if needed.

```
# Indices of variables used for refinement (see grDef.h for definitions)  
EMDA_REFINE_VARIABLE_INDICES = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14,  
15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34,  
35]
```

```
# Number of variables used for refinement
EMDA_NUM_REFINE_VARS = 36
```

we can choose not to test every evolution variable when deciding whether to refine the grid. These parameters control which variables are used to trigger refinement.

```
# CFL (Courant--Friedrichs--Lewy) factor
EMDA_CFL_FACTOR = 0.25
```

```
# Simulation start time
EMDA_RK_TIME_BEGIN = 0
# Simulation end time
EMDA_RK_TIME_END = 1000
```

These parameters control the CFL factor and the total coordinate time for which the simulation is run. The remaining Runge–Kutta settings are typically fine at their default values.

```
#####
# EMDA THEORY PARAMETERS
#####
EMDA_LAMBDA = [1, 1, 1, 1]
EMDA_ALPHA_THEORY = [1.0, 1.0]
EMDA_LF = [1.0, 0.0]
EMDA_P_EXPO = -1.0
EMDA_ALPHA_POS_NEG_FLOOR = false
EMDA_ETA = [0.4, 0.4]
EMDA_XI = [0, 0, 0]
```

```

CHI_FLOOR = 0.000001
EMDA_TRK0 = 0.0
EMDA_ETA_R0 = 1.31
EMDA_ETA_POWER = [2.0, 2.0]
EMDA_ETADAMP = 1.0

```

```

ETA_CONST = 2.0
ETA_R0 = 30.0
ETA_DAMPING = 1.0
ETA_DAMPING_EXP = 1.0

```

These parameters define the physics of the system, many of which control the gauge choice. The two parameters that we modify are α_0 and α_1 , which select the theory being evolved: $\alpha_0 = \alpha_1 = 1$ corresponds to EMDA, $\alpha_0 = 1, \alpha_1 = 0$ corresponds to EMD, and $\alpha_0 = \alpha_1 = 0$ corresponds to Einstein–Maxwell. The remaining gauge parameters are generally fine at their default values.

```

# Kreiss--Oliger dissipation parameter
KO DISS SIGMA = 0.4

```

This parameter controls the strength of numerical dissipation. Some dissipation is required for stability, and a value of 0.4 is a reasonable starting point.

```

#####
# Select initial data type
# See the bottom of the file for reference

EMDA_ID_TYPE = 0

```

This selects the type of initial data used. A value of 0 uses the TPID solver, though other black hole initial-data options are available.

```
#####
```

```
# Set up EMDA grid extents
```

```
EMDA_GRID_MIN_X = -400.0
```

```
EMDA_GRID_MAX_X = 400.0
```

```
EMDA_GRID_MIN_Y = -400.0
```

```
EMDA_GRID_MAX_Y = 400.0
```

```
EMDA_GRID_MIN_Z = -400.0
```

```
EMDA_GRID_MAX_Z = 400.0
```

These parameters define the physical size of the computational grid. The values shown here are reasonable defaults. We do not want the boundaries to close to where we extract the waves as reflections from the boundaries can contaminate our waveforms.

```
#####
```

```
# Black hole parameters
```

```
EMDA_BH1_AMR_R = 1.0
```

```
EMDA_BH1_CONSTRAINT_R = 2.0
```

```
EMDA_BH1_MAX_LEV = 15
```

```
EMDA_BH1 = {MASS = 0.5, X = 5.0, Y = 0.0, Z = 0.0,
```

```
            V_X = -0.00100282, V_Y = 0.095775311, V_Z = 0.0,
```

```
            SPIN = 0.0, SPIN_THETA = 0.0, SPIN_PHI = 0.0,
```

```
            ELECTRIC_CHARGE = 0.05, MAGNETIC_CHARGE = 0.0}
```

```

EMDA_BH2_AMR_R = 1.0
EMDA_BH2_CONSTRAINT_R = 2.0
EMDA_BH2_MAX_LEV = 15
EMDA_BH2 = {MASS = 0.5, X = -5.0, Y = 0.0, Z = 0.0,
            V_X = 0.00100282, V_Y = -0.095775311, V_Z = 0.0,
            SPIN = 0.0, SPIN_THETA = 0.0, SPIN_PHI = 0.0,
            ELECTRIC_CHARGE = 0.05, MAGNETIC_CHARGE = 0.0}

```

Here we set the parameters for each black hole. `EMDA_BH1_AMR_R` defines the radius within which refinement is forced around each black hole, while `EMDA_BH1_CONSTRAINT_R` specifies the radius excluded when computing Hamiltonian and momentum constraints. `EMDA_BH2_MAX_LEV` sets the maximum refinement level near the black hole.

Note that the `X`, `Y`, and `Z` values do not determine the initial placement of the black holes on the grid. Instead, they specify where the code begins searching for the black hole location and define the excision region. For equal-mass binaries ($q = 1$), these should match $\pm b$ in the TPID parameters.

The electric charge must remain smaller than the mass to avoid exceeding the extremal limit. Magnetic charge should not be used yet, as it is still under development. The spin parameter corresponds to $S = \chi m^2$. TPID reads the charge and spin from this line, but not the mass or position.

Note that `V_X`, `V_Y`, and `V_Z` represent the *momenta*, not the velocities. This is an important distinction.

```

#####
# GRAVITATIONAL WAVE EXTRACTION PARAMETERS

```

```

EMDA_GW_NUM_RADAI = 6
EMDA_GW_RADAI = [50.0, 60.0, 70.0, 80.0, 90.0, 100.0]

```

```

EMDA_GW_NUM_LMODES = 7
EMDA_GW_L_MODES = [2, 3, 4, 5, 6, 7, 8]

# Radius at which conserved quantities are extracted
EMDA_GW_EXTRACTION_RADIUS = 300.0

# Modes for EM, dilaton, and axion radiation
EMDA_RAD_NUM_LMODES = 6
EMDA_RAD_L_MODES = [1, 2, 3, 4, 5, 6]
#####

```

These parameters control the extraction of gravitational, electromagnetic, dilaton, and axion radiation. We can choose which spherical harmonics we want to extract. `EMDA_GW_NUM_LMODES` applies to gravitational waves, while `EMDA_RAD_NUM_LMODES` applies to all other radiative fields. The mode numbers correspond to the multipole moments being extracted. Although modes can be defined up to $l \approx 10$, in practice the dominant contributions are usually from modes with $l \leq 4$.

Radiation is extracted on spherical surfaces at several radii, which are specified above.

```

# Time steps between apparent-horizon (AEH) solves
AEH_SOLVER_FREQ = 50

```

This parameter controls how often the apparent-horizon solver is invoked. The remaining AEH-related settings should generally be left at their default values.

```

TPID_PAR_B = 5.0
TPID_TARGET_M_PLUS = 0.48594
TPID_TARGET_M_MINUS = 0.48594

```

TPID_PAR_B sets the coordinate separation between the black holes. The target masses are the values used internally by the TPID solver.

TPID_CENTER_OFFSET = {X = 0.0, Y = 0.0, Z = 0.0}

For equal-mass binaries ($q = 1$), this should be left unchanged. For unequal mass ratios, the system should be recentered on the center of mass.

TPID_NPOINTS_A = 40

TPID_NPOINTS_B = 40

TPID_NPOINTS_PHI = 20

These parameters control the resolution of the TPID grid. The values shown here are reasonable defaults.

TPID_GIVE_BARE_MASS = 1

This determines whether the puncture masses are fixed or solved for. Setting this to 0 instructs TPID to solve for the masses. When set to 1, TPID attempts to match the puncture masses to the specified target values.

A recommended procedure is to first estimate the expected masses (e.g., $m = 0.5$ for $q = 1$), run TPID with mass solving enabled, then update the target masses and rerun with TPID_GIVE_BARE_MASS = 1 to finalize the initial data.

TPID_REPLACE_LAPSE_WITH_SQRT_CHI = true

This option should always be enabled when using SSL.

Include dilatonic contributions in rho during TPID solve

TPID_USE_DILATON = true

```
# Interpolated initial data should not be used with TPID_USE_DILATON
# You should also be using the EMD equations in this case
TPID_USE_INTERPOLATED_SOLUTION = false
```

If both options are set to `false`, TPID will generate pure Einstein–Maxwell initial data. Setting `TPID_USE_INTERPOLATED_SOLUTION = true` produces EMD initial data, while `TPID_USE_DILATON = true` produces EMMA initial data.

You should ensure that the α parameters are consistent with the type of initial data being generated, and that CMake is configured to build the corresponding system of equations.

Appendix G

ParaView

This chapter provides a brief introduction to some of the basic functions in ParaView. We begin by loading simulation output files and then discuss how to select the appropriate ParaView data file.

G.1 Opening a File in ParaView

To load data into ParaView, click the **Open File** button in the top-left toolbar. This button is highlighted in Figure G.1.

After clicking the **Open File** button, navigate to the directory where your data is saved. Dendro commonly outputs data in `.vtu` format, which stands for *VTK Unstructured Grid*. For parallel runs, the output is often organized using a `.pvtu` file. The `.pvtu` file acts as a wrapper that tells ParaView how the individual `.vtu` pieces fit together.

In most cases, you should open only the `.pvtu` file rather than opening the individual `.vtu` files by hand.

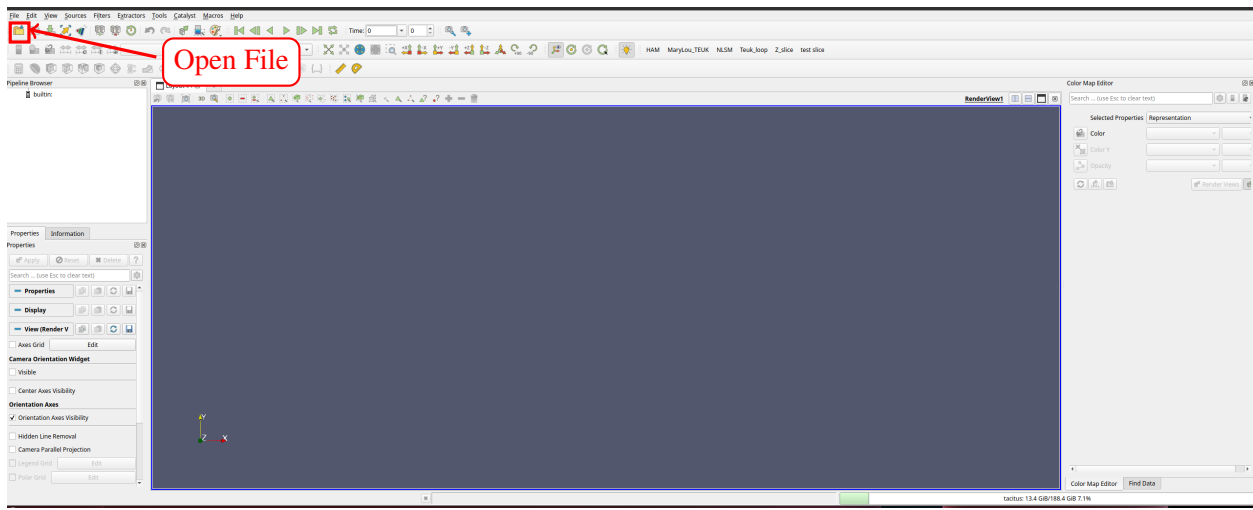


Figure G.1 The ParaView interface with the **Open File** icon highlighted.

G.2 Creating a Slice

Once the data has been loaded, a common operation is to create a two-dimensional slice through the three-dimensional dataset. In this example, the **Z_slice** button is used to create a slice through the data, as shown in Figure G.2.

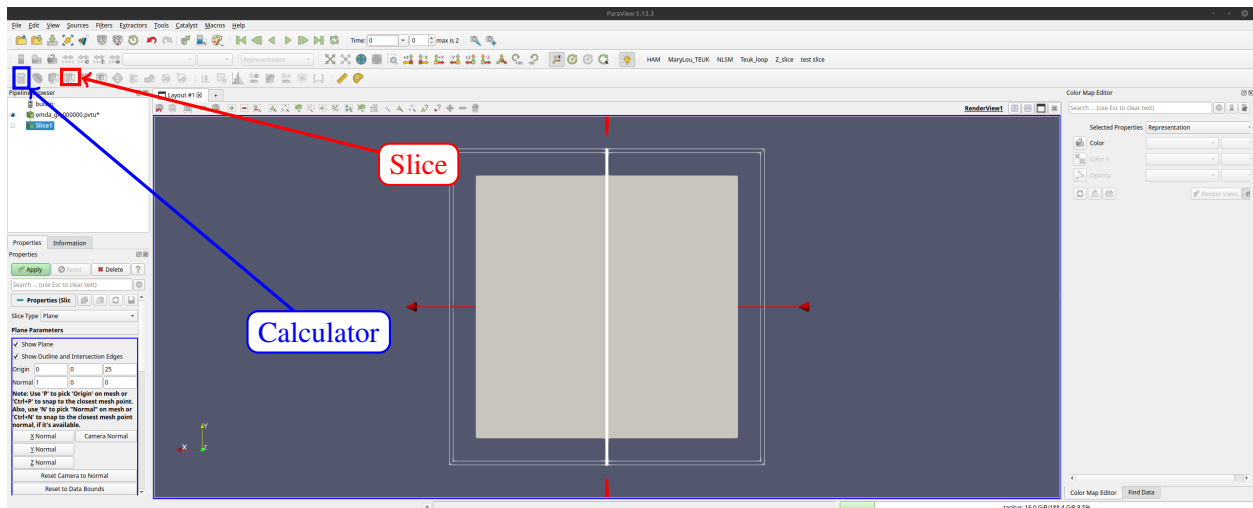


Figure G.2 The ParaView interface with the **Z_slice** button highlighted.

G.3 Adjusting the Slice and Viewing the Data and the Calculator

After selecting the **Slice** tool, ParaView allows us to choose how the data should be sliced. Often, we want to create z -slices, since this is the default orientation used by Dendro output. To do this, set the slice normal to $(0, 0, 1)$. We also set the origin to $(0, 0, 10^{-5})$. The small nonzero z -value is used because setting $z = 0$ can sometimes cause ParaView to miss the slice.

The **Show Plane** option should be turned off so that only the data slice is displayed. After changing these settings, click **Apply** again.

Once the slice is displayed, we can change which field is visualized using the field-selection menu. We can also move forward and backward through the available time steps using the time controls. Finally, the representation menu allows us to switch between different viewing modes, such as **Surface** and **Wireframe**. These controls are highlighted in Figure G.3.

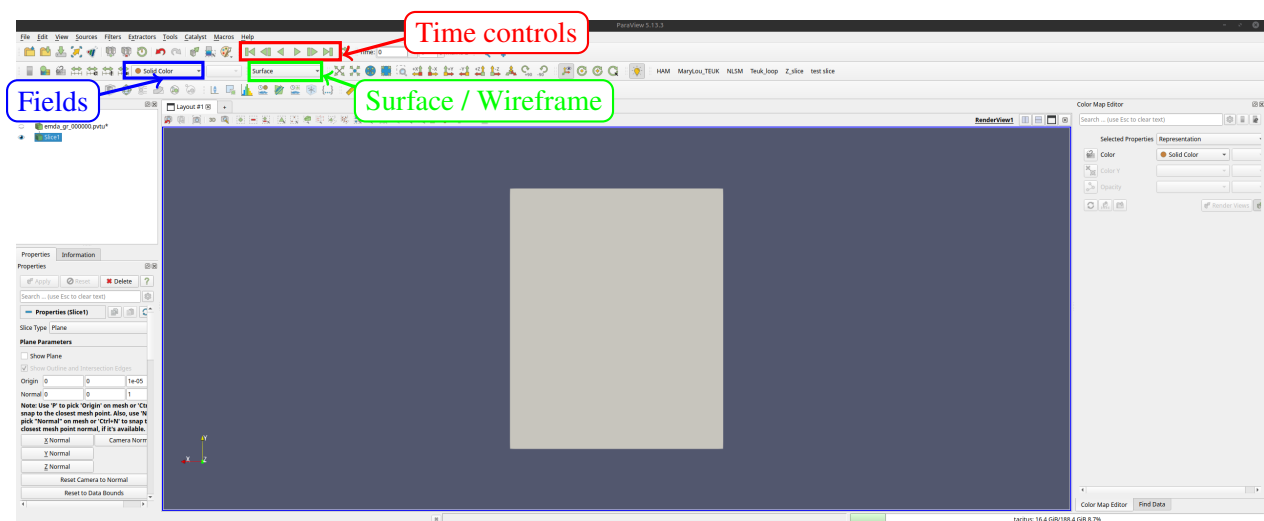


Figure G.3 ParaView controls for stepping through time, selecting the displayed field or color option, and switching between representations such as **Surface** and **Wireframe**.

The Calculator tool is opened by clicking the Calculator button. In the Calculator window, the available fields can be loaded by selecting Scalars. Mathematical operations may then be applied to these fields to construct derived quantities.

G.4 Warp by Scalar

To use the **Warp By Scalar** filter, click **Filters** and search for **Warp By Scalar**. After selecting the filter, its settings will appear in the Properties panel, in the same general location as the slice settings.

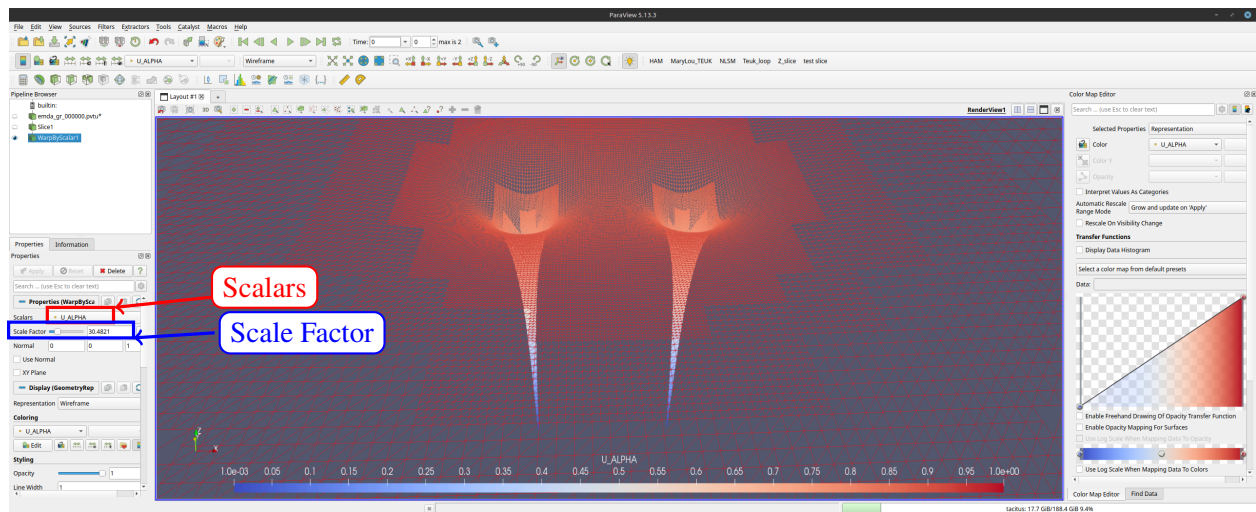


Figure G.4 The **Warp By Scalar** settings in the Properties panel. The **Scalars** menu selects the field used for the warping, and the **Scale Factor** controls the strength of the warp.

G.5 Color Map Editor

The **Color Map Editor** controls how ParaView maps data values to colors. It can be used to change the color scale, adjust the visible data range, and choose a different color preset.

By default, the color mapping may not automatically update when stepping through time. The easiest way to ensure that the colors match the data range at each time step is to enable **Clamp** and update the color range at every time step. Otherwise, the coloring may not match the current range of the data unless it is updated manually.

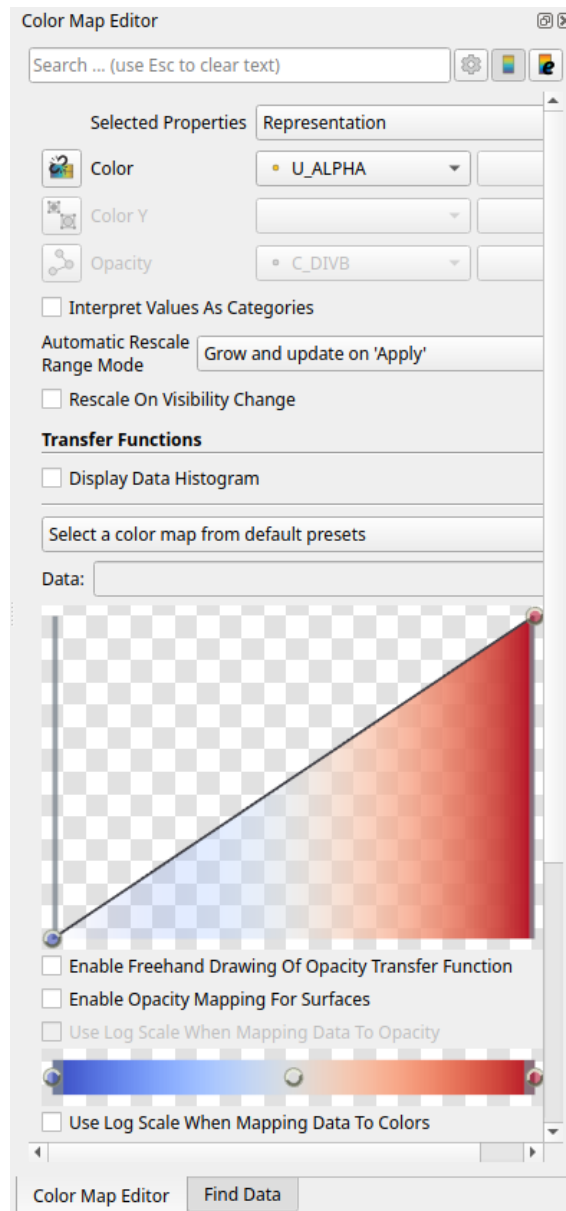


Figure G.5 The **Color Map Editor** panel in ParaView. This panel controls the color scale, opacity, and transfer function used to visualize the selected field.

G.6 Macros, making Animations Saving CSV Files, and Finding Data

Often, we want to perform repetitive tasks in ParaView. These tasks can be automated by creating a macro. To create a macro, click **Tools**, then select **Start Trace**. Perform the task you want to automate, and when you are finished, click **Stop Trace**. ParaView will then allow you to save the recorded trace as a macro. The macro can later be used to repeat the same sequence of actions automatically.

We can also save the data currently loaded in ParaView as a CSV file. To do this, click **File**, then select **Save Data**. This will save the data from the currently active object, such as the current slice. The resulting CSV file can then be loaded into Python, MATLAB, Mathematica, or another tool for later analysis.

Animations can also be saved from ParaView by clicking **File** and then selecting **Save Animation**.

Finally, ParaView can be used to find data that satisfies certain conditions. To do this, click **View** and then select **Find Data**. In the **Find Data** panel, choose the dataset you want to search. You can usually leave the element type set to **Point**. Then choose the field you want to search, set the desired condition and value, and click **Find Data**. The matching region will be highlighted in the render view.

Appendix H

General Notes

H.1 Equations of Motion

We start with the equations of motion for EMDA. We begin with the action in the Einstein frame:

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{8\pi G} - 2(\nabla\phi)^2 - e^{-2\alpha_0\phi} F^2 - \frac{1}{2} e^{4\alpha_1\phi} (\nabla\kappa)^2 - \kappa F_{ab} (*F)^{ab} \right]. \quad (\text{H.1})$$

Here, κ is the axion, related to the three-form potential by $H_{abc} = \frac{1}{2} e^{4\phi} \epsilon_{abc}{}^d \partial_d \kappa$, and α_0 and α_1 parameterize the theory. In this instance, we are interested in the sector of low-energy string theory where $\alpha_0 = \alpha_1 = 1$.

The covariant equations of motion are:

$$0 = \nabla^a \nabla_a \phi + \frac{1}{2} \alpha_0 e^{-2\alpha_0\phi} F^2 - \frac{1}{2} \alpha_1 e^{4\alpha_1\phi} (\nabla\kappa)^2, \quad (\text{H.2})$$

$$0 = \nabla^a \left(e^{4\alpha_1\phi} \nabla_a \kappa \right) - F_{ab} (*F)^{ab}, \quad (\text{H.3})$$

$$0 = \nabla_a \left(e^{-2\alpha_0\phi} F^{ab} \right) + (*F)^{ab} \nabla_a \kappa. \quad (\text{H.4})$$

The Einstein equation takes the form

$$R_{ab} - \frac{1}{2} g_{ab} R = 8\pi G \left[2 \left(\nabla_a \phi \nabla_b \phi - \frac{1}{2} g_{ab} (\nabla\phi)^2 \right) + 2e^{-2\alpha_0\phi} \left(F_{ac} F_b{}^c - \frac{1}{4} g_{ab} F^2 \right) + \frac{1}{2} e^{4\alpha_1\phi} \left(\nabla_a \kappa \nabla_b \kappa - \frac{1}{2} g_{ab} (\nabla\kappa)^2 \right) \right]. \quad (\text{H.5})$$

To the Maxwell constraints we add constraint damping so that violations propagate away at the speed of light. In particular, we introduce two scalar fields Φ and Ψ :

$$\begin{aligned} \nabla^b (F_{ba} + g_{ba} \Psi) &= \kappa_1 n_a \Psi + 2\alpha_0 \nabla^b \phi F_{ba} \\ &\quad + e^{2\alpha_0\phi} (*F)_{ab} \nabla_b \kappa, \end{aligned} \quad (\text{H.6})$$

$$\nabla^b ((*F)_{ba} + g_{ba} \Phi) = \kappa_2 n_a \Phi. \quad (\text{H.7})$$

For the gravitational part we have

$$(\partial_t - \mathcal{L}_\beta) \tilde{\gamma}_{ij} = -2\alpha \tilde{A}_{ij}, \quad (\text{H.8})$$

$$(\partial_t - \mathcal{L}_\beta) \chi = \frac{2}{3} \alpha \chi K, \quad (\text{H.9})$$

$$(\partial_t - \mathcal{L}_\beta) K = [\dots] + 4\pi\alpha (\rho + T), \quad (\text{H.10})$$

$$(\partial_t - \mathcal{L}_\beta) \tilde{A}_{ij} = [\dots] - 8\pi\alpha \left(\chi T_{ij} - \frac{T}{3} \tilde{\gamma}_{ij} \right), \quad (\text{H.11})$$

$$(\partial_t - \mathcal{L}_\beta) \tilde{\Gamma}^i = [\dots] - 16\pi\alpha \chi^{-1} J^i, \quad (\text{H.12})$$

$$\tilde{\Gamma}^i = \tilde{\gamma}^{jk} \tilde{\Gamma}^i_{jk}. \quad (\text{H.13})$$

Here $[\dots]$ denotes the standard right-hand side of the BSSN equations in the absence of source terms.

The quantities T_{ij} , T , J^i , and ρ are defined as

$$\begin{aligned} J^i &\equiv -h^{ia} T_{ab} n^b \\ &= \frac{1}{8\pi} (2\Pi D^i \phi + 2e^{-2\alpha_0 \phi} \epsilon^{ijk} E_j B_k + \frac{1}{2} e^{4\alpha_1 \phi} \Xi D^i \kappa) \end{aligned} \quad (\text{H.14})$$

$$\begin{aligned} T_{ij} &\equiv T_{ab} h^a_i h^b_j \\ &= \frac{1}{8\pi} \left[2 \left[D_i \phi D_j \phi - \frac{1}{2} h_{ij} ((D\phi)^2 - \Pi^2) \right] \right. \\ &\quad \left. - 2e^{-2\alpha_0 \phi} \left[E_i E_j + B_i B_j - \frac{1}{2} h_{ij} (E_k E^k + B_k B^k) \right] \right. \\ &\quad \left. + \frac{1}{2} e^{4\alpha_1 \phi} \left[D_i \kappa D_j \kappa - \frac{1}{2} h_{ij} ((D\kappa)^2 - \Xi^2) \right] \right]. \end{aligned} \quad (\text{H.15})$$

$$T = T^{ij} \gamma_{ij}, \quad (\text{H.16})$$

$$\begin{aligned} \rho &\equiv \frac{1}{8\pi} \left(\Pi^2 + (D\phi)^2 + e^{-2\alpha_0 \phi} (E_a E^a + B_a B^a) \right. \\ &\quad \left. + \frac{1}{4} e^{4\alpha_1 \phi} (\Xi^2 + (D\kappa)^2) \right) \end{aligned} \quad (\text{H.17})$$

For the matter equations we have:

$$(\partial_t - \mathcal{L}_\beta) \phi = -\alpha \Pi. \quad (\text{H.18})$$

$$(\partial_t - \mathcal{L}_\beta)\kappa = -\alpha\Xi. \quad (\text{H.19})$$

$$\begin{aligned} (\partial_t - \mathcal{L}_\beta)\Pi &= \alpha K\Pi \\ &\quad - \alpha\chi \tilde{\gamma}^{ij} \partial_i \partial_j \phi - \chi \tilde{\gamma}^{ij} \partial_i \alpha \partial_j \phi \\ &\quad + \alpha\chi \tilde{\Gamma}^i \partial_i \phi + \frac{\alpha}{2} \tilde{\gamma}^{ij} \partial_i \chi \partial_j \phi \\ &\quad - \frac{\alpha\alpha_0}{\chi} e^{-2\alpha_0\phi} \tilde{\gamma}_{ij} B^i B^j + \frac{\alpha\alpha_0}{\chi} e^{-2\alpha_0\phi} \tilde{\gamma}_{ij} E^i E^j \\ &\quad + \frac{\alpha\alpha_1}{2} e^{4\alpha_1\phi} \chi \tilde{\gamma}^{ij} \partial_i \kappa \partial_j \kappa - \frac{\alpha\alpha_1}{2} e^{4\alpha_1\phi} \Xi^2. \end{aligned} \quad (\text{H.20})$$

$$\begin{aligned} (\partial_t - \mathcal{L}_\beta)\Xi &= \alpha K\Xi \\ &\quad - \alpha\chi \tilde{\gamma}^{ij} \partial_i \partial_j \kappa - \chi \tilde{\gamma}^{ij} \partial_i \alpha \partial_j \kappa \\ &\quad + \alpha\chi \tilde{\Gamma}^i \partial_i \kappa + \frac{\alpha}{2} \tilde{\gamma}^{ij} \partial_i \kappa \partial_j \chi \\ &\quad - 4\alpha\alpha_1 \chi \tilde{\gamma}^{ij} \partial_i \kappa \partial_j \phi + 4\alpha\alpha_1 \Pi\Xi \\ &\quad + \frac{4\alpha}{\chi} e^{-4\alpha_1\phi} \tilde{\gamma}_{ij} B^i E^j. \end{aligned} \quad (\text{H.21})$$

$$\begin{aligned} (\partial_t - \mathcal{L}_\beta)\Psi &= -\alpha \partial_i E^i \\ &\quad + \frac{3\alpha}{2\chi} E^i \partial_i \chi + 2\alpha_0 \alpha E^i \partial_i \phi \\ &\quad + \alpha e^{2\alpha_0\phi} B^i \partial_i \kappa - \alpha\eta_1 \Psi. \end{aligned} \quad (\text{H.22})$$

$$\begin{aligned} (\partial_t - \mathcal{L}_\beta)\Phi &= \alpha \partial_i B^i \\ &\quad - \frac{3\alpha}{2\chi} B^i \partial_i \chi - \alpha\eta_2 \Phi. \end{aligned} \quad (\text{H.23})$$

$$\begin{aligned}
(\partial_t - \mathcal{L}_\beta)E^i &= \alpha K E^i \\
&+ 2\alpha\alpha_0\chi^{1/2}\epsilon^{jik}\tilde{\gamma}_{kl}B^l\partial_j\phi - 2\alpha\alpha_0E^i\Pi \\
&+ \alpha e^{2\alpha_0\phi}\chi^{1/2}\epsilon^{ijk}\tilde{\gamma}_{kl}E^l\partial_j\kappa - \alpha e^{2\alpha_0\phi}\Xi B^i \\
&- \alpha\chi\tilde{\gamma}^{ij}\partial_j\Psi - \chi^{1/2}\epsilon^{jik}\tilde{\gamma}_{kl}B^l\partial_j\alpha \\
&- \alpha\chi^{1/2}\epsilon^{jik}B^l\partial_j\tilde{\gamma}_{kl} - \alpha\chi^{1/2}\epsilon^{jik}\tilde{\gamma}_{kl}\partial_jB^l \\
&+ \alpha\chi^{-1/2}\epsilon^{jik}\tilde{\gamma}_{kl}B^l\partial_j\chi.
\end{aligned} \tag{H.24}$$

$$\begin{aligned}
(\partial_t - \mathcal{L}_\beta)B^i &= \alpha K B^i + \alpha\chi\tilde{\gamma}^{ij}\partial_j\Phi \\
&- \chi^{1/2}\epsilon^{ijk}\tilde{\gamma}_{kl}E^l\partial_j\alpha + \alpha\chi^{-1/2}\epsilon^{ijk}\tilde{\gamma}_{kl}E^l\partial_j\chi \\
&- \alpha\chi^{1/2}\epsilon^{ijk}E^l\partial_j\tilde{\gamma}_{kl} - \alpha\chi^{1/2}\epsilon^{ijk}\tilde{\gamma}_{kl}\partial_jE^l.
\end{aligned} \tag{H.25}$$

We also make use of the slow-start lapse proposed in [62]. Using this approach reduces the amount of computational resources spent refining junk radiation.

H.2 Initial Data

As a first step, we assume conformally flat space; that is, we decompose the metric as

$$\gamma_{ij} = \psi^4 \tilde{\gamma}_{ij}, \tag{H.26}$$

where a tilde denotes conformal variables and ψ is the conformal factor. The assumption of conformal flatness simplifies the following computations and is a reasonable choice for the systems we study. However, Kerr spacetimes are known not to admit conformally flat slices [63, 64]. Alternative approaches have been proposed to construct initial data for highly spinning black holes [65].

Notwithstanding this, Bowen–York black holes can reach spin values of order $\chi \sim 0.9$ [66]. Since our simulations remain below this limit, the approximation should be acceptable.

We nevertheless anticipate the appearance of “junk radiation” at the beginning of the simulation, as has been observed in vacuum GR [67] and in charged cases [61]. This spurious radiation decays as the system relaxes to quasi-equilibrium. We adopt maximal slicing conditions, $K = 0$. Using a conformally rescaled extrinsic curvature, we can express it in terms of a vector V^i as

$$\tilde{A}^{ij} = 2\delta^{ik}\delta^{jh}V_{(h,k)} - \frac{2}{3}\delta^{ij}\partial_k V^k. \quad (\text{H.27})$$

Under these assumptions, the problem reduces to determining the vector V^i and the conformal factor ψ to fully specify γ_{ij} and $\tilde{A}_{ij}$. The constraint equations take the form

$$\nabla^2\psi + \frac{1}{8}\psi^{-7}\tilde{A}_{ij}\tilde{A}^{ij} + 2\pi\psi^5\rho = 0, \quad (\text{H.28})$$

$$(\nabla^2 V)^i + \frac{1}{3}\delta^{ij}\partial_j(\partial_k V^k) - 8\pi J^i = 0. \quad (\text{H.29})$$

We also impose the Gauss and no-monopole constraints. Following [53, 57], we rescale the electromagnetic fields as

$$\tilde{E}^i = \psi^6 E^i, \quad \tilde{B}^i = \psi^6 B^i, \quad (\text{H.30})$$

which makes the Maxwell constraints

$$\partial_i \tilde{E}^i = 0, \quad \partial_i \tilde{B}^i = 0, \quad (\text{H.31})$$

independent of ψ and therefore solvable separately from the spacetime variables. Since the equations are linear, solutions may be superimposed. Following [58], we neglect contributions from the magnetic field.

The momentum constraint is linear and does not depend on ψ , allowing us to decompose the solution into vacuum GR and matter contributions:

$$V^i = V_{\text{GR}}^i + V_{\text{M}}^i. \quad (\text{H.32})$$

For V_{GR}^i , we use the standard Bowen–York solutions [68]. For the matter contribution, we solve

$$\nabla^2 V^i + \frac{1}{3} \delta^{ij} \partial_j (\partial_k V^k) - 8\pi J^i = 0, \quad (\text{H.33})$$

where

$$J^i = 2\Pi D^i \phi + 2e^{-2\alpha_0 \phi} \epsilon^{ijk} E_j B_k + \frac{1}{2} e^{4\alpha_1 \phi} \Xi D^i \kappa. \quad (\text{H.34})$$

Assuming maximal slicing, time symmetry, and neglecting contributions from the magnetic field and axion, we obtain $J^i = 0$. Thus $V_{\text{M}}^i = 0$, and the momentum constraints reduce to the standard Bowen–York form.

The Hamiltonian constraint becomes

$$\nabla^2 \psi + \frac{1}{8} \psi^{-7} \tilde{A}_{ij} \tilde{A}^{ij} + 2\pi \psi^5 \rho = 0. \quad (\text{H.35})$$

Separating the singular part of ψ from the finite correction u yields

$$\psi = \sqrt{\left(1 + u + \sum_{n=1}^{N_p} \frac{M_n}{2R_n}\right)^2 - \left(\sum_{n=1}^{N_p} \frac{Q_n}{2R_n}\right)^2}. \quad (\text{H.36})$$

This corresponds to a Reissner–Nordström black hole plus corrections. Following [58], we define

$$\eta = \sum_{n=1}^{N_p} \frac{M_n}{2R_n}, \quad \xi = \sum_{n=1}^{N_p} \frac{Q_n}{2R_n}, \quad \kappa = 1 + u + \eta, \quad (\text{H.37})$$

so that the conformal factor can be written as

$$\psi = \sqrt{\eta^2 - \xi^2}. \quad (\text{H.38})$$

Rearranging the Hamiltonian constraint gives

$$\begin{aligned} \nabla^2 u = -\frac{1}{\kappa} & \left(\partial_a \kappa \partial^a \kappa - \partial_a \xi \partial^a \xi - \partial_a \psi \partial^a \psi \right. \\ & \left. + \frac{1}{8} \psi^{-6} \tilde{A}_{ij} \tilde{A}^{ij} + 2\pi \psi^5 \rho \right), \quad (\text{H.39}) \end{aligned}$$

with

$$\begin{aligned} \rho \equiv & \Pi^2 + (D\phi)^2 + e^{-2\alpha_0\phi} (E_a E^a + B_a B^a) \\ & + \frac{1}{4} e^{4\alpha_1\phi} (\Xi^2 + (D\kappa)^2). \end{aligned} \quad (\text{H.40})$$

Under the same simplifying assumptions as before, ρ reduces to

$$\rho \rightarrow (D\phi)^2 + e^{-2\alpha_0\phi} (E_a E^a). \quad (\text{H.41})$$

This follows [58] closely, except that we include dilaton contributions in ρ . Specifically, we employ a Reissner–Nordström electric field for the electric component and add a dilaton field for a single, spinless black hole. The axion is initialized to zero and allowed to relax to a quasi-equilibrium state during evolution.

The electric and dilaton fields are defined as:

$$M_i \equiv \text{physical mass of puncture } i, \quad (\text{H.42})$$

$$q_i \equiv \text{electric charge of puncture } i, \quad (\text{H.43})$$

$$m_i \equiv M_i - \frac{q_i^2}{2M_i}, \quad (\text{H.44})$$

$$\alpha_i \equiv \tanh^{-1} \left(\frac{q_i}{M_i \sqrt{2}} \right), \quad (\text{H.45})$$

$$f_i(r_i) \equiv \left(r_i + \frac{1}{2} m_i \right)^2, \quad (\text{H.46})$$

$$\bar{\rho}_i^2(r_i) \equiv f_i(r_i)^2 + 2m_i f_i(r_i) r_i \sinh^2(\alpha_i), \quad (\text{H.47})$$

$$\Phi_i(r_i) \equiv -\frac{1}{2} \ln \left(\frac{\bar{\rho}_i^2(r_i)}{f_i(r_i)^2} \right), \quad (\text{H.48})$$

$$\Phi(r) \equiv \sum_{i=1}^{N_p} \Phi_i(r_i). \quad (\text{H.49})$$

and

$$E^i = \psi^{-6} \sum_{i=1}^{N_p} \frac{q_i}{r_i}. \quad (\text{H.50})$$

Perhaps future approaches making use of hyperbolic constraint formulations, such as those proposed in [69–72], could be adapted to produce improved initial data in the future.

H.3 Wave Extractions

The gravitational wave signals are given by the Newman–Penrose scalar derived from the Weyl tensor as:

$$\Psi_4 = C_{\alpha\beta\gamma\delta} k^\alpha \bar{m}^\beta k^\gamma \bar{m}^\delta. \quad (\text{H.51})$$

Here, k and $\bar{m}$ are part of a null tetrad, which we define below. The extraction of the radiated electromagnetic energy is given by two scalar quantities calculated in the Newman–Penrose formalism [56, 73, 74]. The triads are constructed to be orthogonal to the normal unit vector n^ν , and they are defined as:

$$l^\alpha = \frac{1}{\sqrt{2}} (n^\alpha + u^\alpha), \quad (\text{H.52})$$

$$k^\alpha = \frac{1}{\sqrt{2}} (n^\alpha - u^\alpha), \quad (\text{H.53})$$

$$m^\alpha = \frac{1}{\sqrt{2}} (v^\alpha - iw^\alpha). \quad (\text{H.54})$$

Asymptotically, the triad vectors u , v , and w behave as unit radial, polar, and azimuthal vectors $\hat{r}$, $\hat{\theta}$, and $\hat{\phi}$, respectively. Similarly, we can construct analogous quantities for the electromagnetic radiation. With this construction, the two radiative quantities are defined as:

$$\xi_1 \equiv \frac{1}{2} F_{\mu\nu} (l^\mu k^\nu + \bar{m}^\mu m^\nu),$$

$$\xi_2 \equiv F_{\mu\nu} \bar{m}^\mu k^\nu.$$

The radiative terms for the dilaton and axion are defined in a similar way. For the dilaton, we have:

$$\Phi = \Pi + r^i D_i \phi, \quad (\text{H.55})$$

and similarly, for the axion, we have:

$$\kappa = \Xi + r^i D_i \kappa, \quad (\text{H.56})$$

where Π and Ξ are the time derivatives of the dilaton and axion, respectively.

At a given extraction radius R_{ex} , we perform a multipolar decomposition by projecting Ψ_4 , ξ_1 , ξ_2 , Φ , and κ onto spherical harmonics of spin weights $s = -2, 0$, and -1 :

$$\Psi_4(t, \theta, \phi) = \sum_{l,m} \psi^{lm}(t) Y_{lm}^{-2}(\theta, \phi), \quad (\text{H.57})$$

$$\xi_1(t, \theta, \phi) = \sum_{l,m} \xi_1^{lm}(t) Y_{lm}^0(\theta, \phi), \quad (\text{H.58})$$

$$\xi_2(t, \theta, \phi) = \sum_{l,m} \xi_2^{lm}(t) Y_{lm}^{-1}(\theta, \phi), \quad (\text{H.59})$$

$$\Phi(t, \theta, \phi) = \sum_{l,m} \Phi^{lm}(t) Y_{lm}^0(\theta, \phi), \quad (\text{H.60})$$

$$\kappa(t, \theta, \phi) = \sum_{l,m} \kappa^{lm}(t) Y_{lm}^0(\theta, \phi). \quad (\text{H.61})$$

We can calculate the radiated energy from the gravitational, electromagnetic, axion, and dilaton fields as follows [57]:

$$F_{\text{GW}} = \frac{dE_{\text{GW}}}{dt} = \lim_{r \rightarrow \infty} \frac{r^2}{16\pi} \sum_{l,m} \left| \int_{-\infty}^t dt' \psi^{lm}(t') \right|^2, \quad (\text{H.62})$$

$$F_{\text{M}} = \frac{dE_{\text{M}}}{dt} = \lim_{r \rightarrow \infty} \frac{r^2}{4\pi} \sum_{l,m} |\phi_2^{lm}(t)|^2. \quad (\text{H.63})$$

H.4 Table of Tests Done

Binary black hole configurations

(a) Kerr–Sen Same-Sign Charge and Spin-Aligned Black Hole Tests

(χ_1, Q_1)	(χ_2, Q_2)	Dilaton ϕ	Axion κ
$\chi_1 = 0.20, Q_1 \in \{0.50, 0.25, 0.125, 0.0624\}$	$\chi_2 \in \{0.20, 0.10, 0.05\}, Q_2 \in \{0.50, 0.25, 0.125, 0.0624\}$	✓	✓
$\chi_1 = 0.10, Q_1 \in \{0.50, 0.25, 0.125, 0.0624\}$	$\chi_2 \in \{0.10, 0.05\}, Q_2$ as above	✓	✓
$\chi_1 = 0.05, Q_1 \in \{0.50, 0.25, 0.125, 0.0624\}$	$\chi_2 = 0.05, Q_2$ as above	✓	✓

(b) Collisions with Other Black Holes

Black Hole 1 (Q, χ)	Black Hole 2 (Q, χ)	ϕ	κ
Kerr–Sen (0.5, 0.2)	Kerr–Sen (0.5, 0.2)	✓	✓
Kerr–Sen (0.5, 0.2)	Kerr (0.0, 0.2)	✓	✓
Kerr–Sen (0.5, 0.2)	Kerr–Sen (−0.5, 0.2)	✗	✗
Kerr–Sen (0.5, −0.2)	Kerr–Sen (0.5, 0.2)	✓	✗
RN (0.5, 0.0)	Kerr (0.0, 0.2)	✓	✓
KN (0.5, 0.2) [†]	KN (0.5, 0.2) [†]	✓	✓

Appendix I

Waveform Catalog

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